Pipelined ADC Concept

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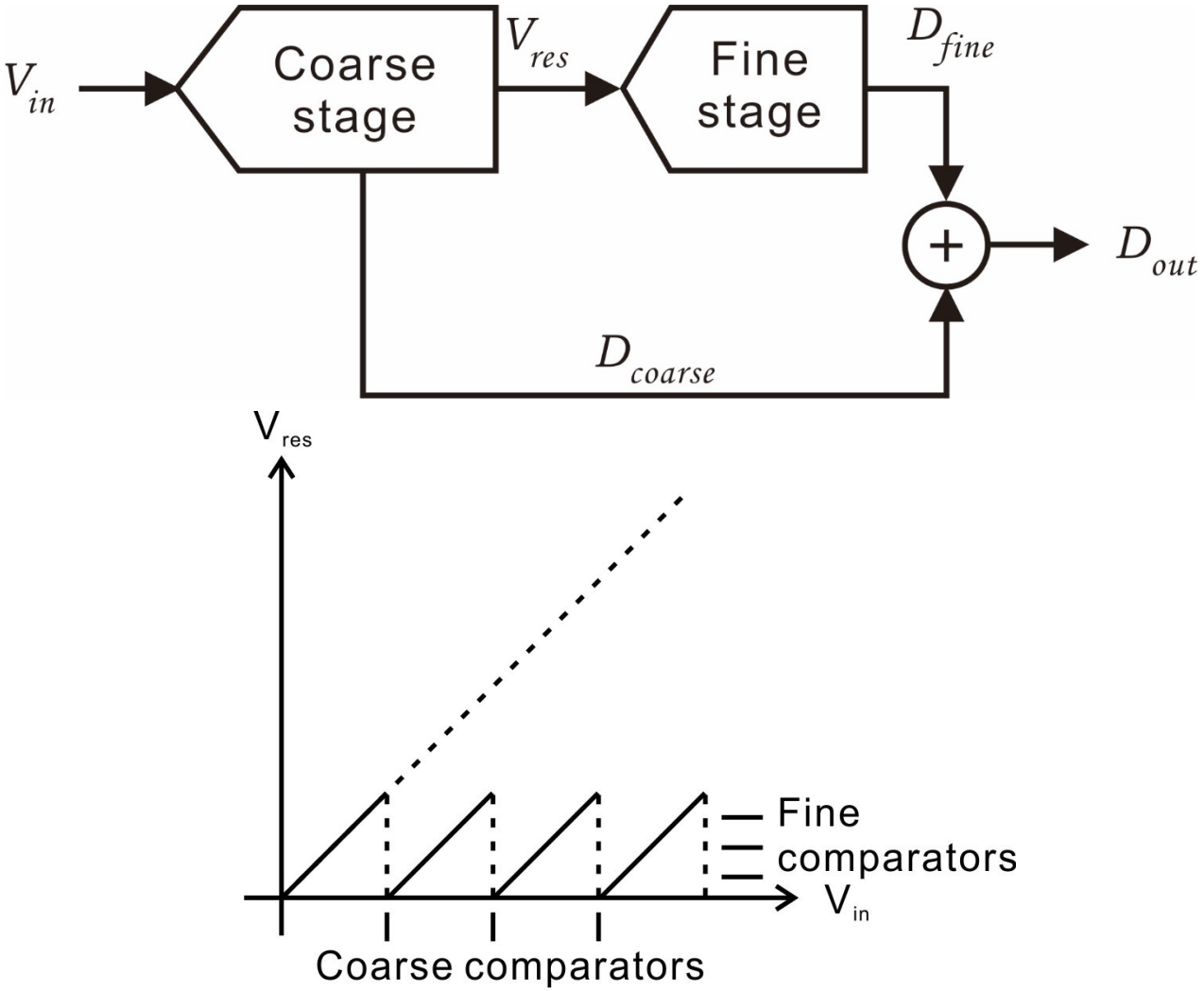
Objective

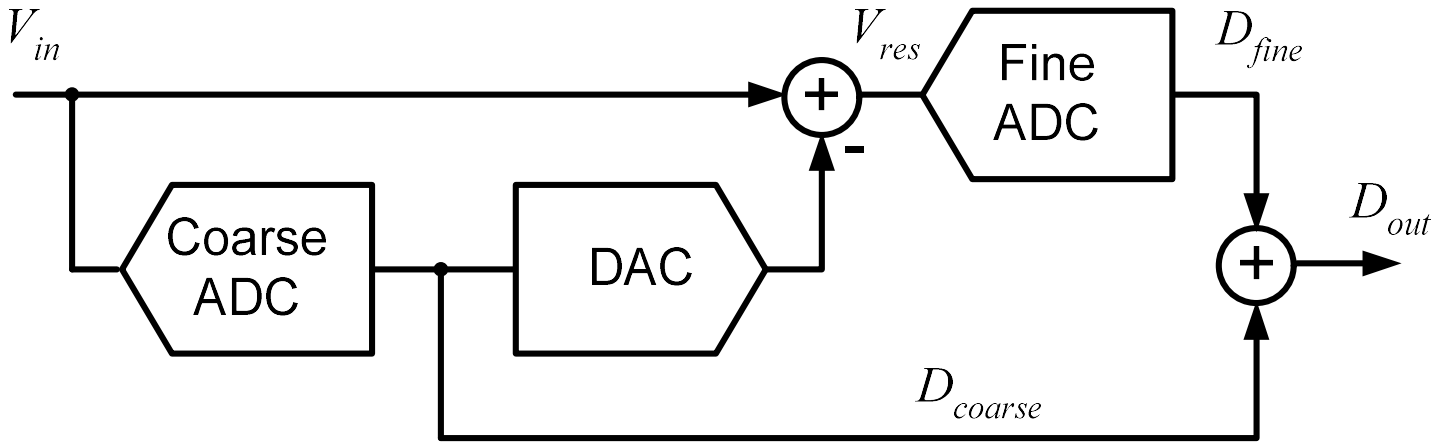
- Want an ADC with
  - High speed
  - High resolution
  - Low hardware complexity
    - Reduce number of comparators
    - Relax the requirement on comparator precision

Similar Idea as Folding ADC

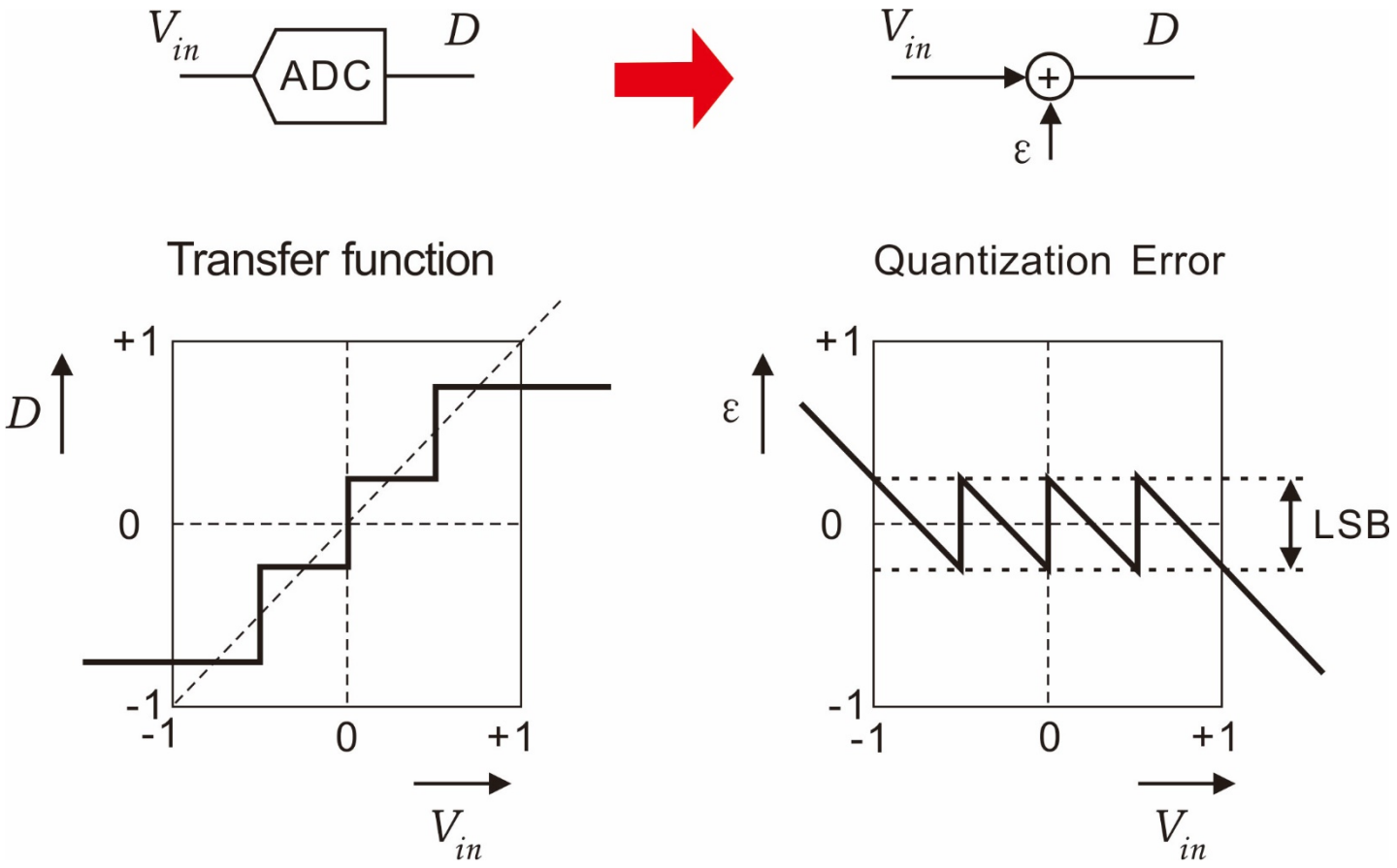


Two-Step Conversion

- General idea
  1. Perform a "coarse" quantization of the input
  2. Compute residual of step 1 conversion using a DAC and a subtractor
  3. Digitize residual using a "fine" quantizer and digitally add to output

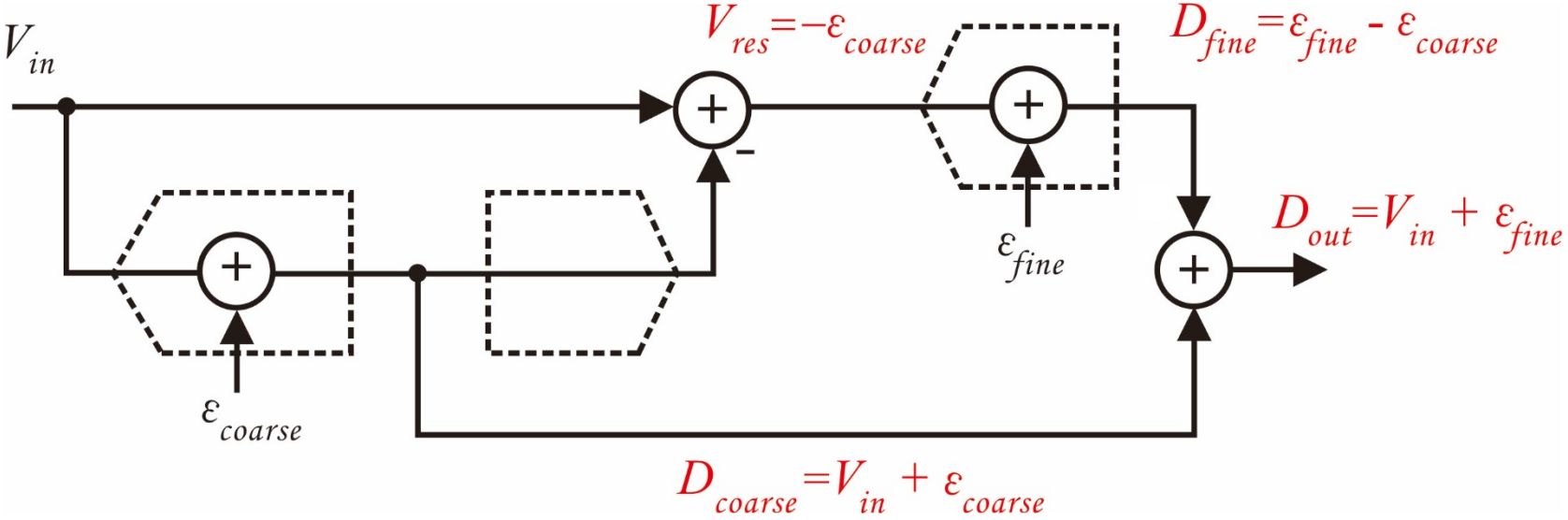


Quantizer Model



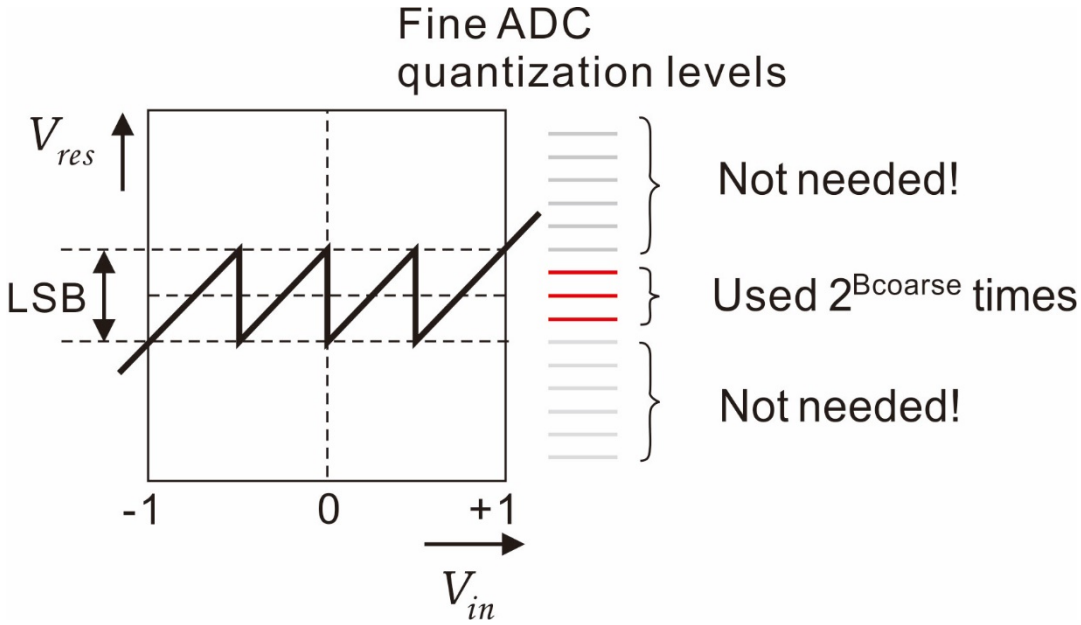
Analysis

- Assuming ideal DAC



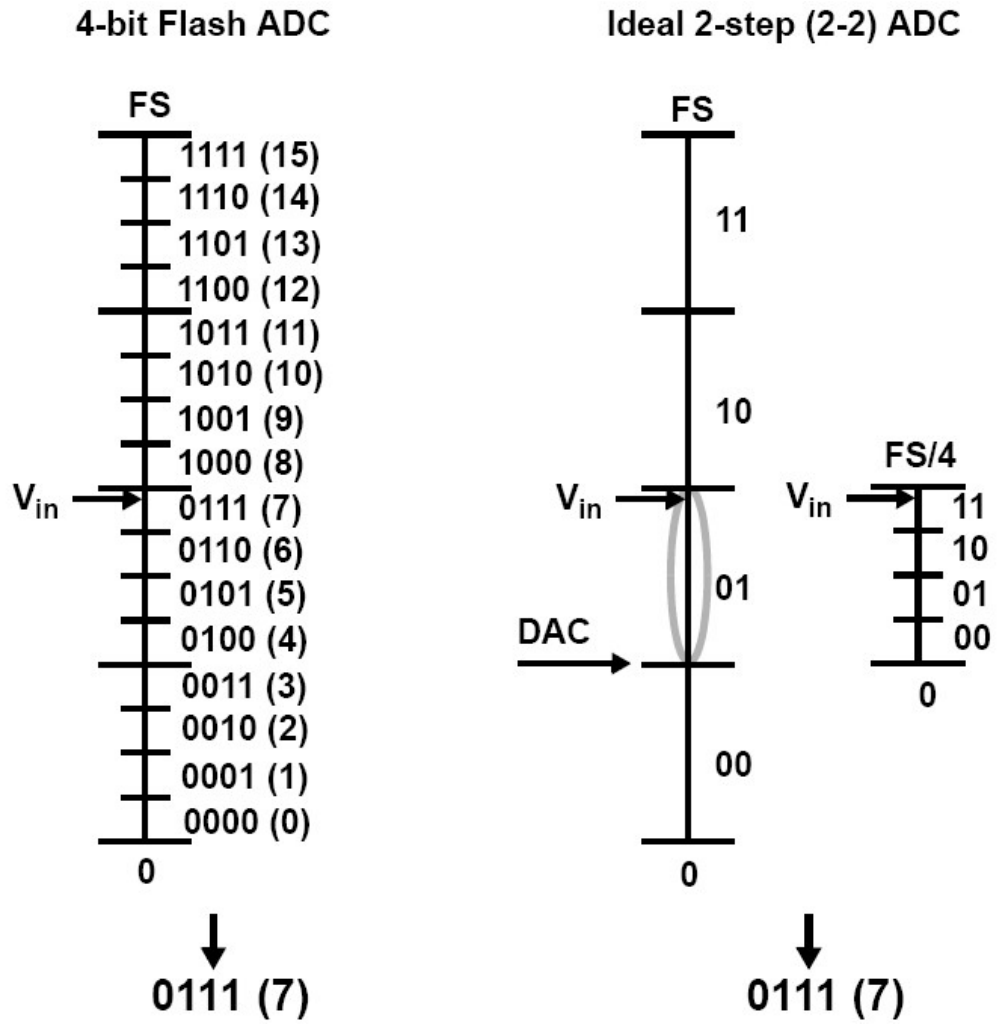
- Output contains only quantization error from fine ADC ☺

Input to Fine Quantizer (V_{res})

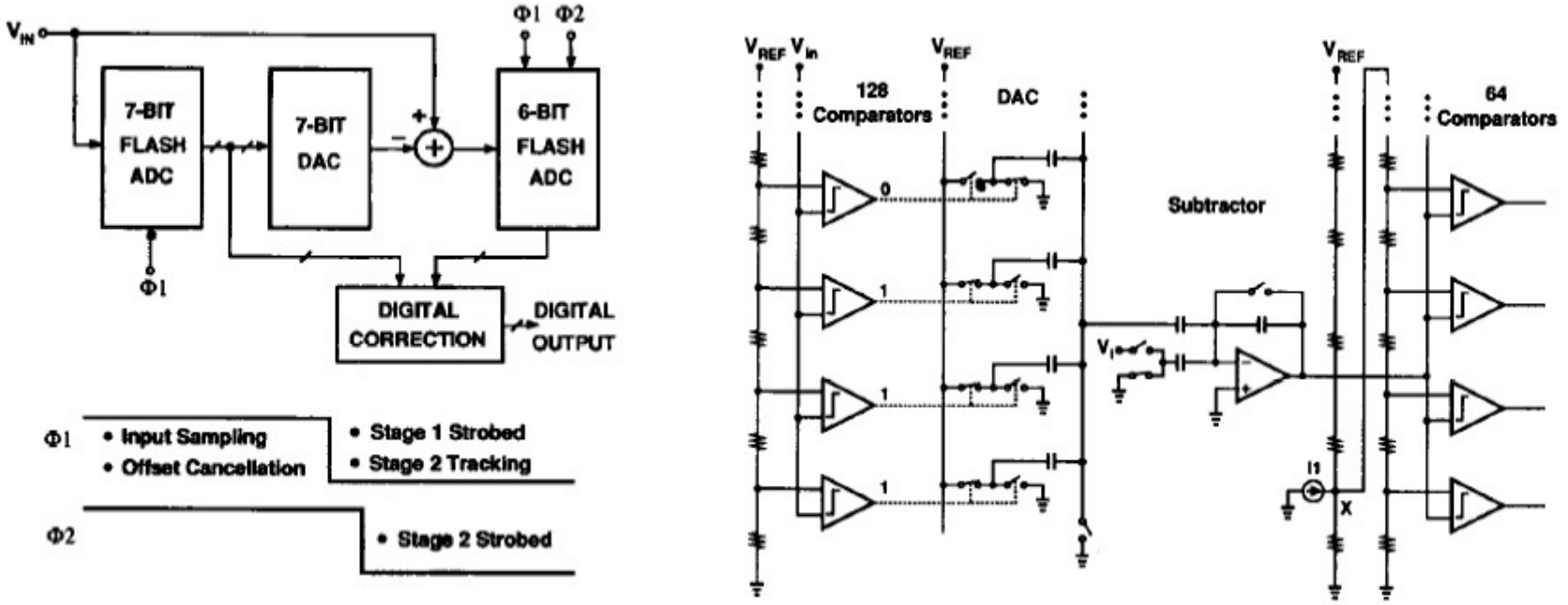


- Example
  - 2b coarse ADC and 2b fine ADC
  - Need only 6 comparators, compared to 15 in a 4-bit flash ADC
    - Advantage becomes more pronounced at higher resolutions

Alternative Illustration

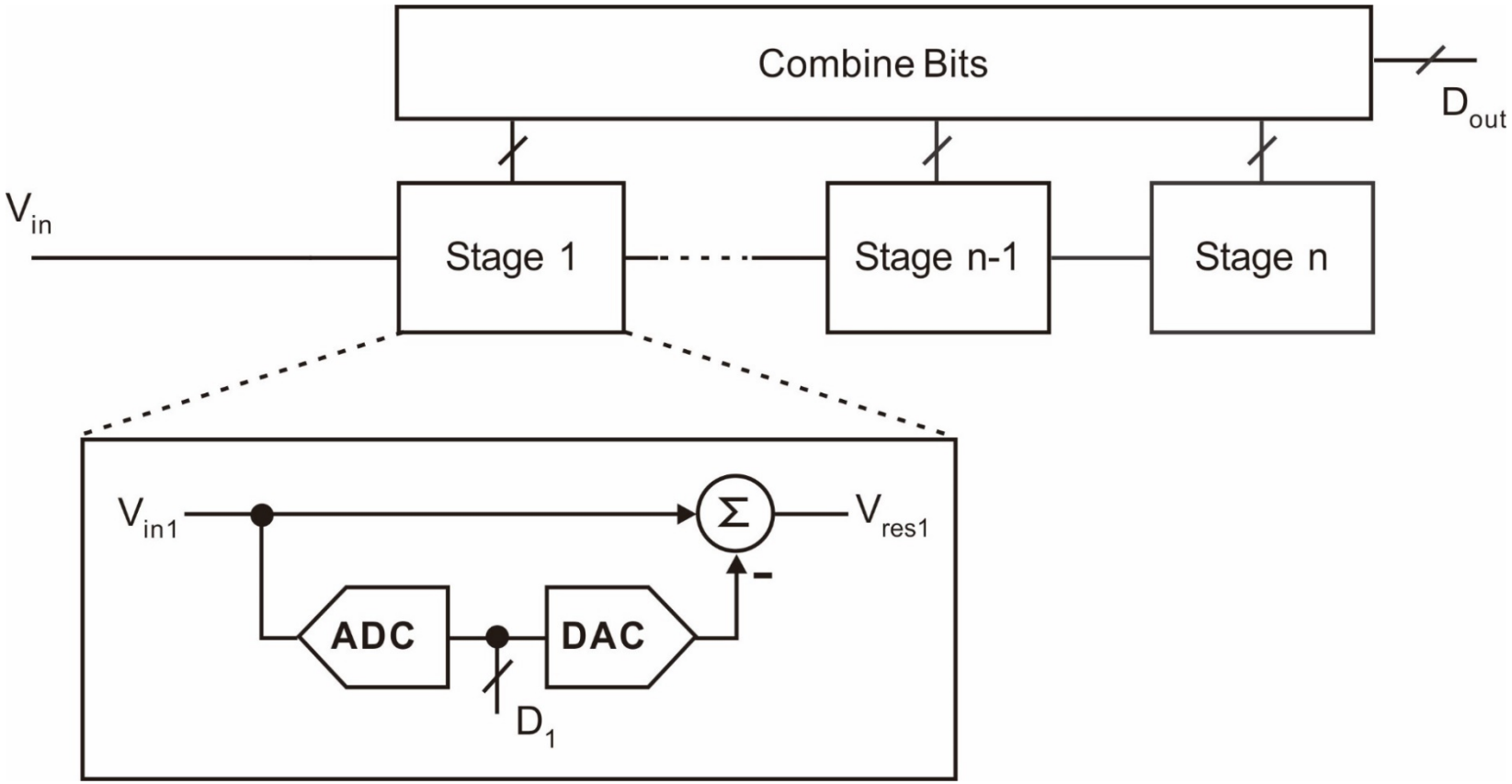


Two-Step ADC Example

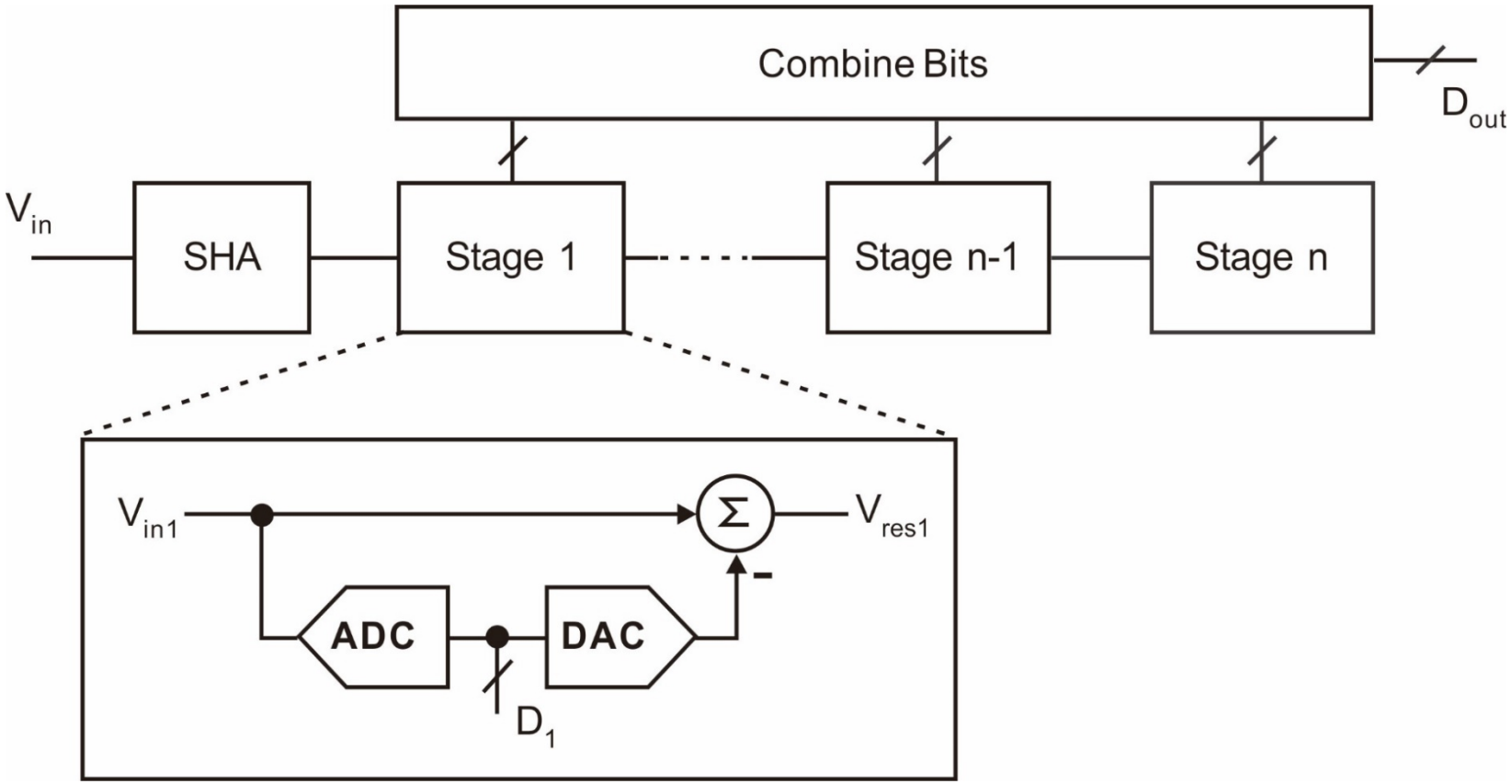


B. Razavi and B.A. Wooley, "A 12-b 5-Msample/s two-step CMOS A/D converter," IEEE JSSC, pp. 1667-1678, Dec. 1992.

Multi-Step ADC

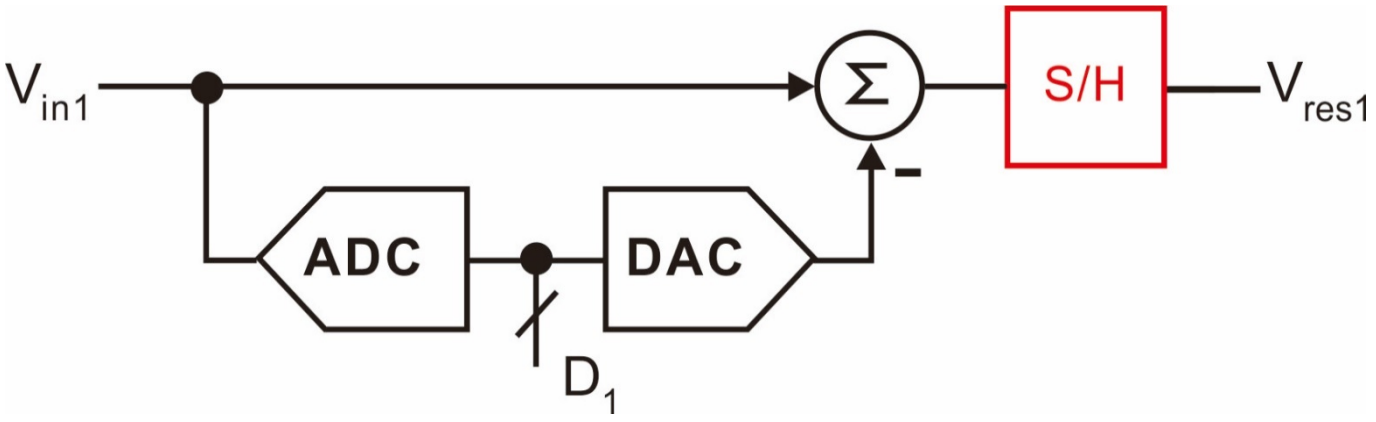


Multi-Step ADC



Increase Conversion Speed

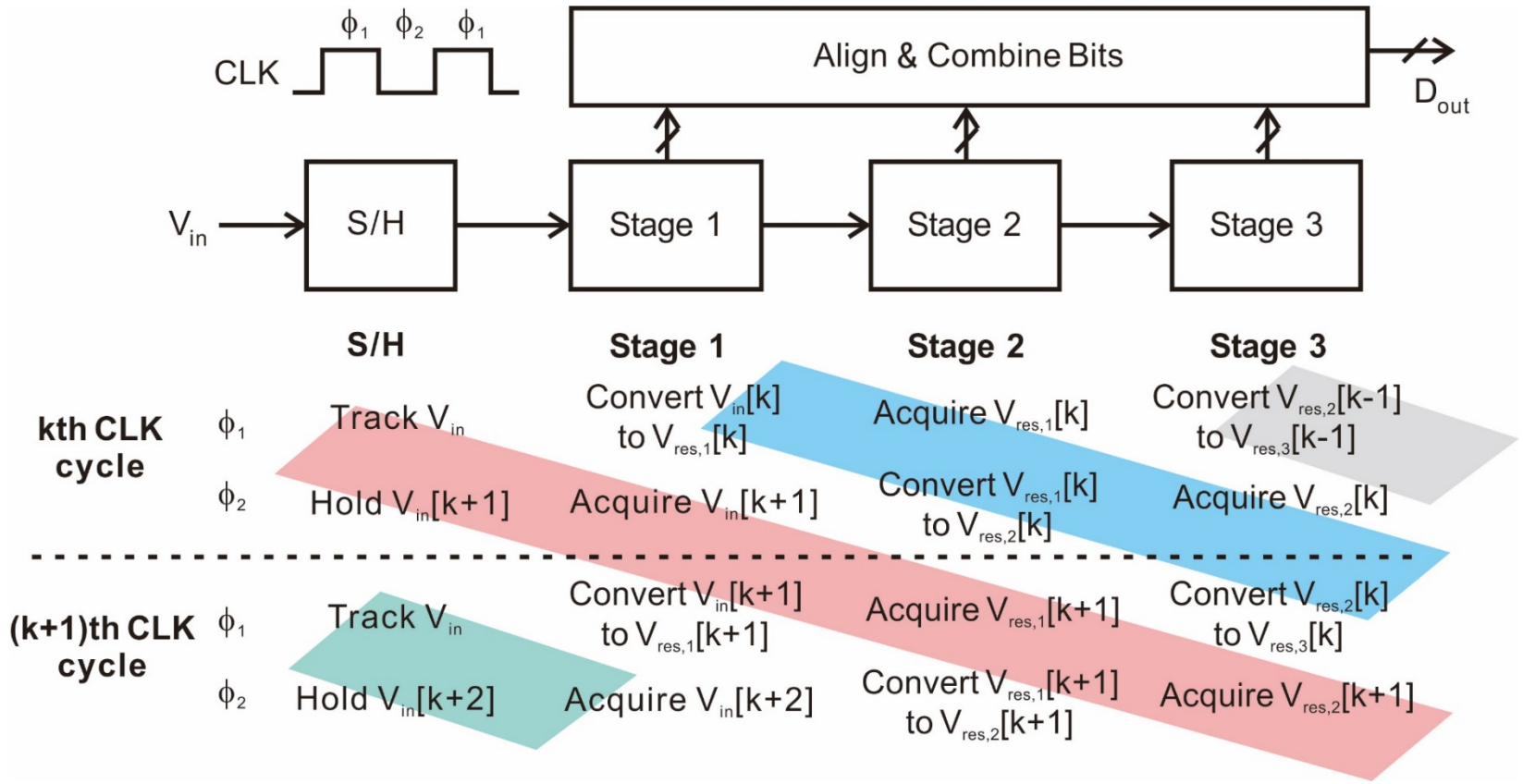
- Long conversion time:
  - $T_{conv} \cong N \cdot (T_{ADC} + T_{DAC} + T_{SUB})$
- Solution
  - Turn delay into latency by pipelining
  - Add a S/H after each stage



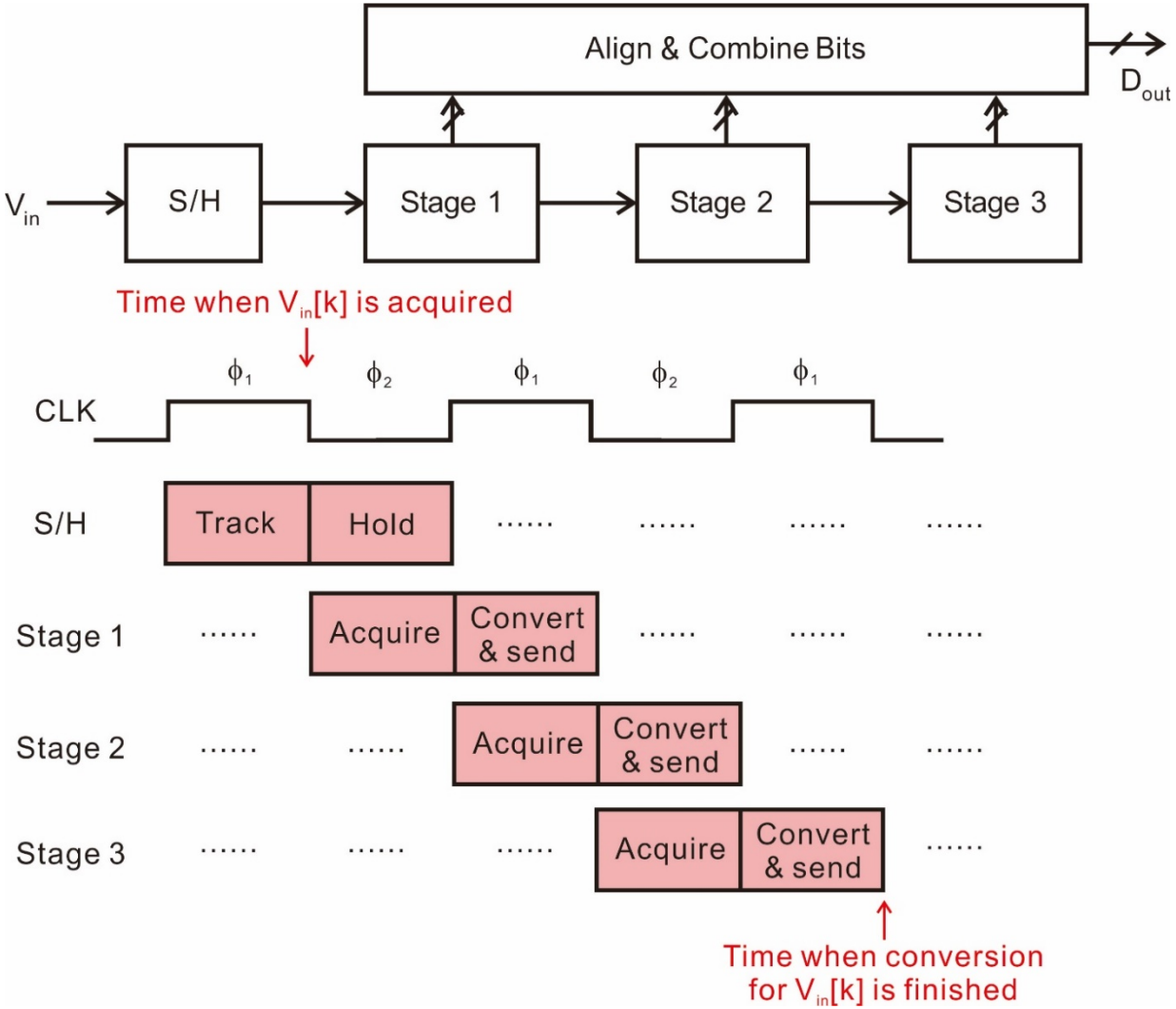
Pipelining – An Old Idea

- At the Ford Motor Company's Highland Park, Mich., plant in 1913

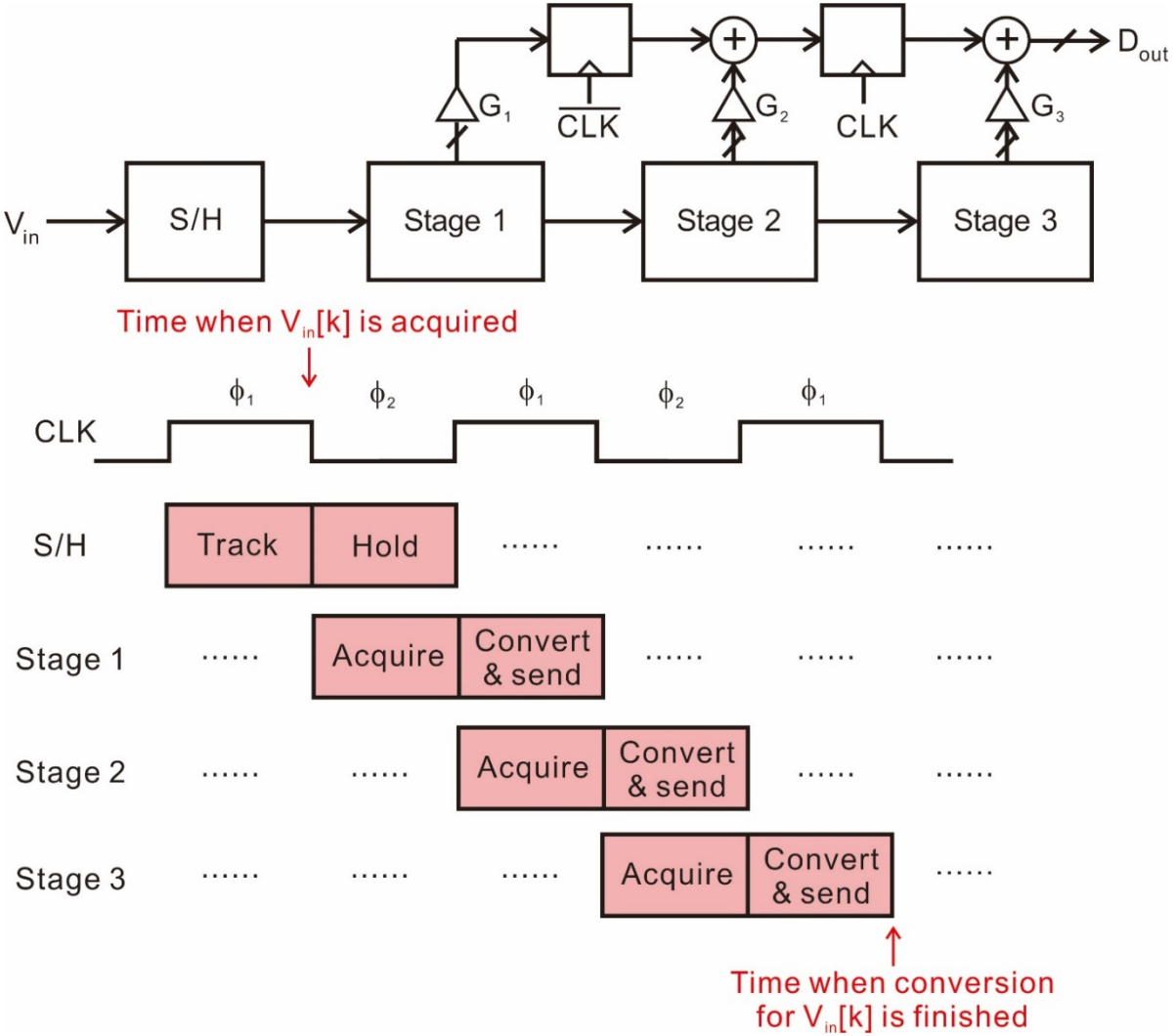
Pipelined Operation (1)



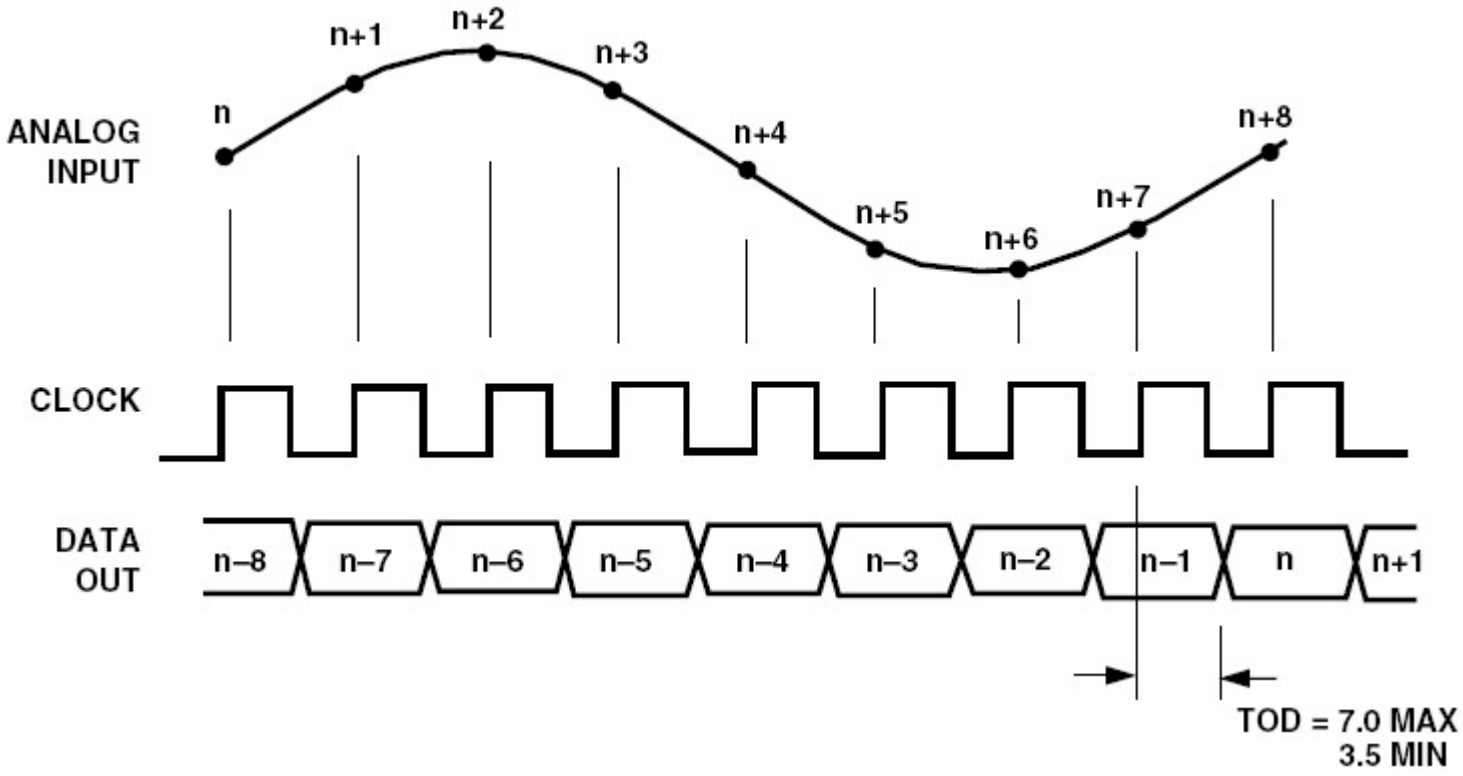
Pipelined Operation (2)



Bit Alignment



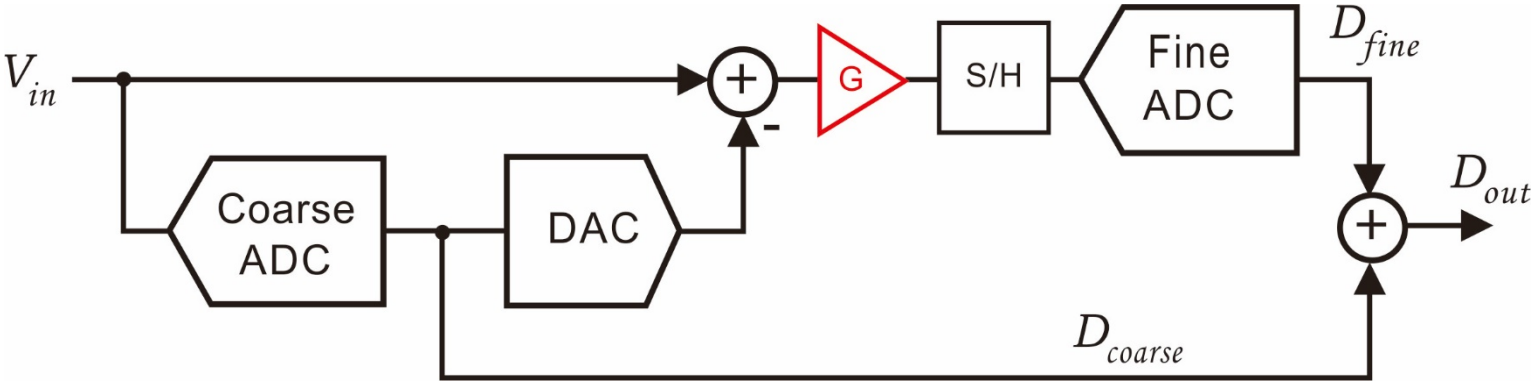
Latency



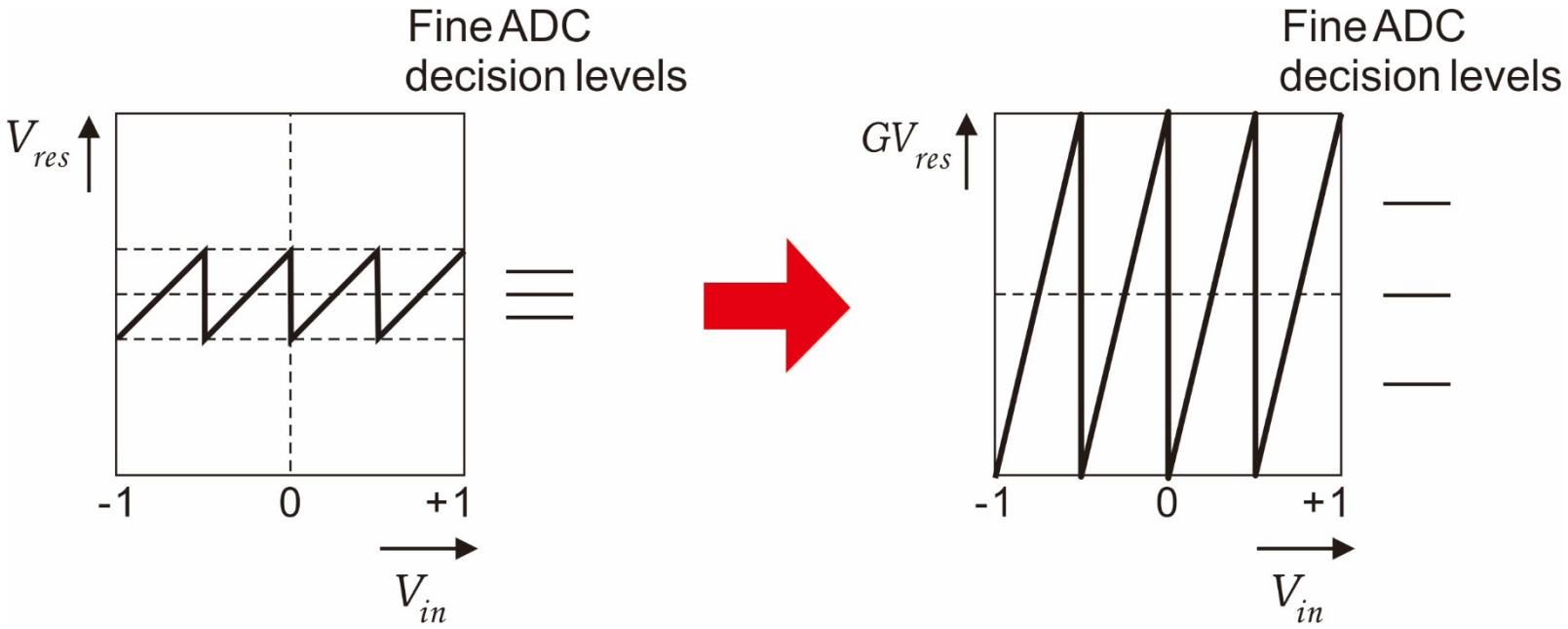
[Analog Devices, AD9226 Data Sheet]

Relax Comparator Requirement

- Fine ADC(s) must have high precision
  - E.g. 8-bit converter with 4-bit/4-bit partition; fine 4-bit decision levels must have "8-bit precision"
- Solution
  - Introduce gain after subtraction

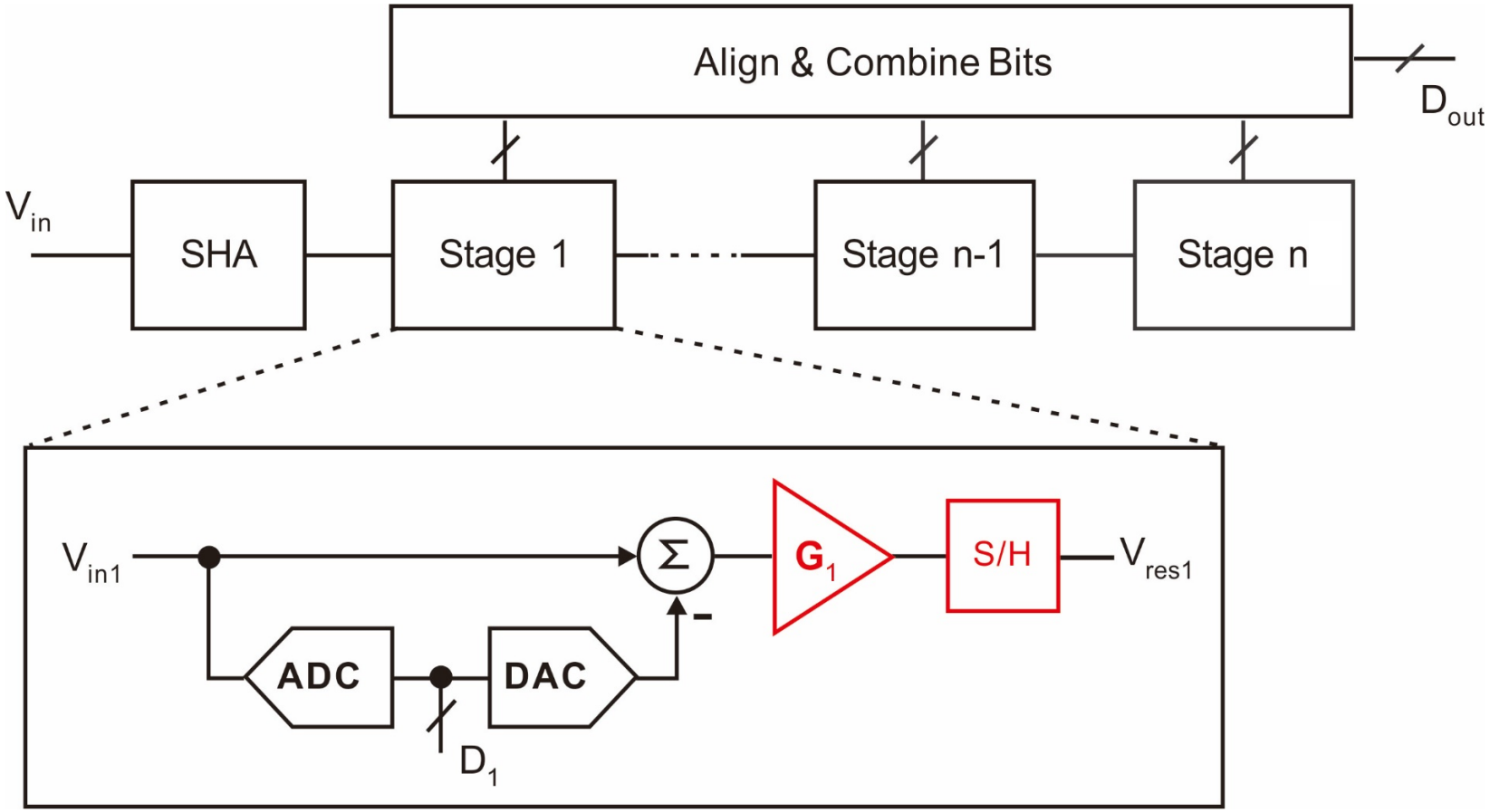


Input to Fine Quantizer with Gain



- No need for precision comparators

Pipeline ADC Block Diagram

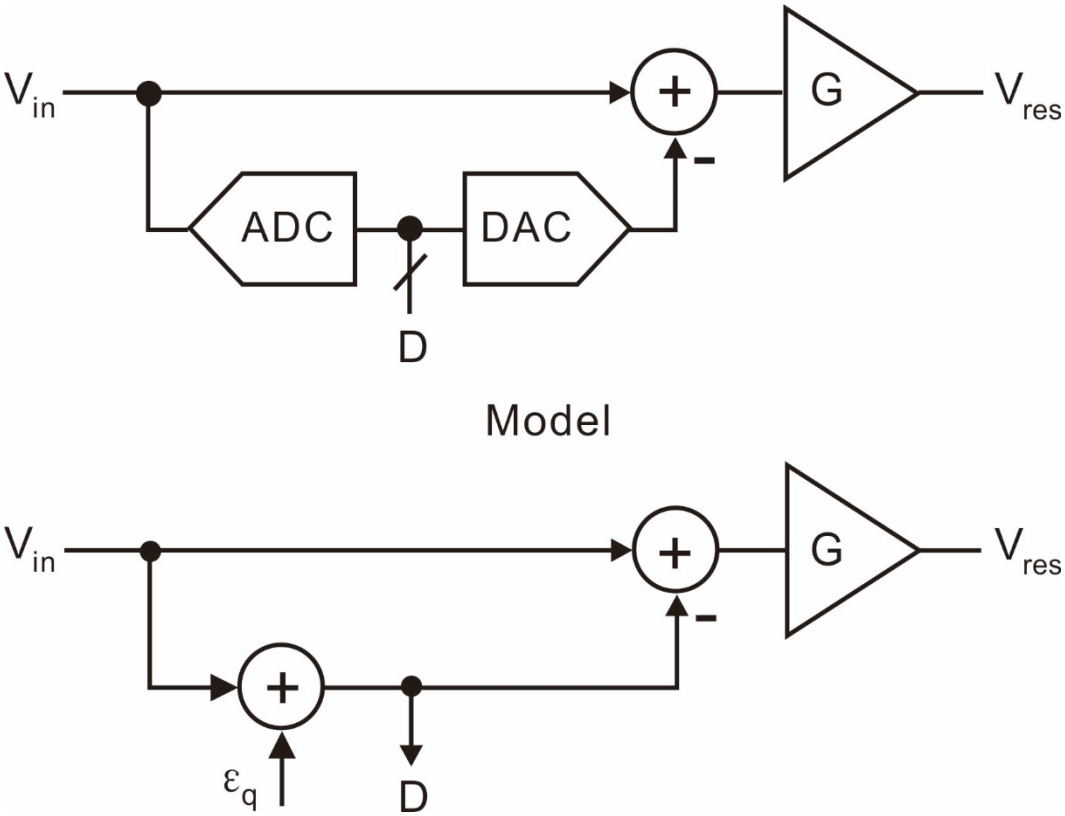


Pipeline ADC Characteristics

- Number of pipelined stages increases linearly with resolution
  - Hardware complexity ~ B
  - Flash ADC hardware complexity ~ 2^B
- Pipeline ADC trades latency for conversion speed
  - Conversion speed limited by only one stage delay
    - Enables high-speed operation
  - Latency can be an issue in some applications
    - E.g. in feedback control loops
- Pipelining only possible with good analog "memory elements"
  - Calls for CMOS implementation using switched-capacitor circuits

Stage Analysis

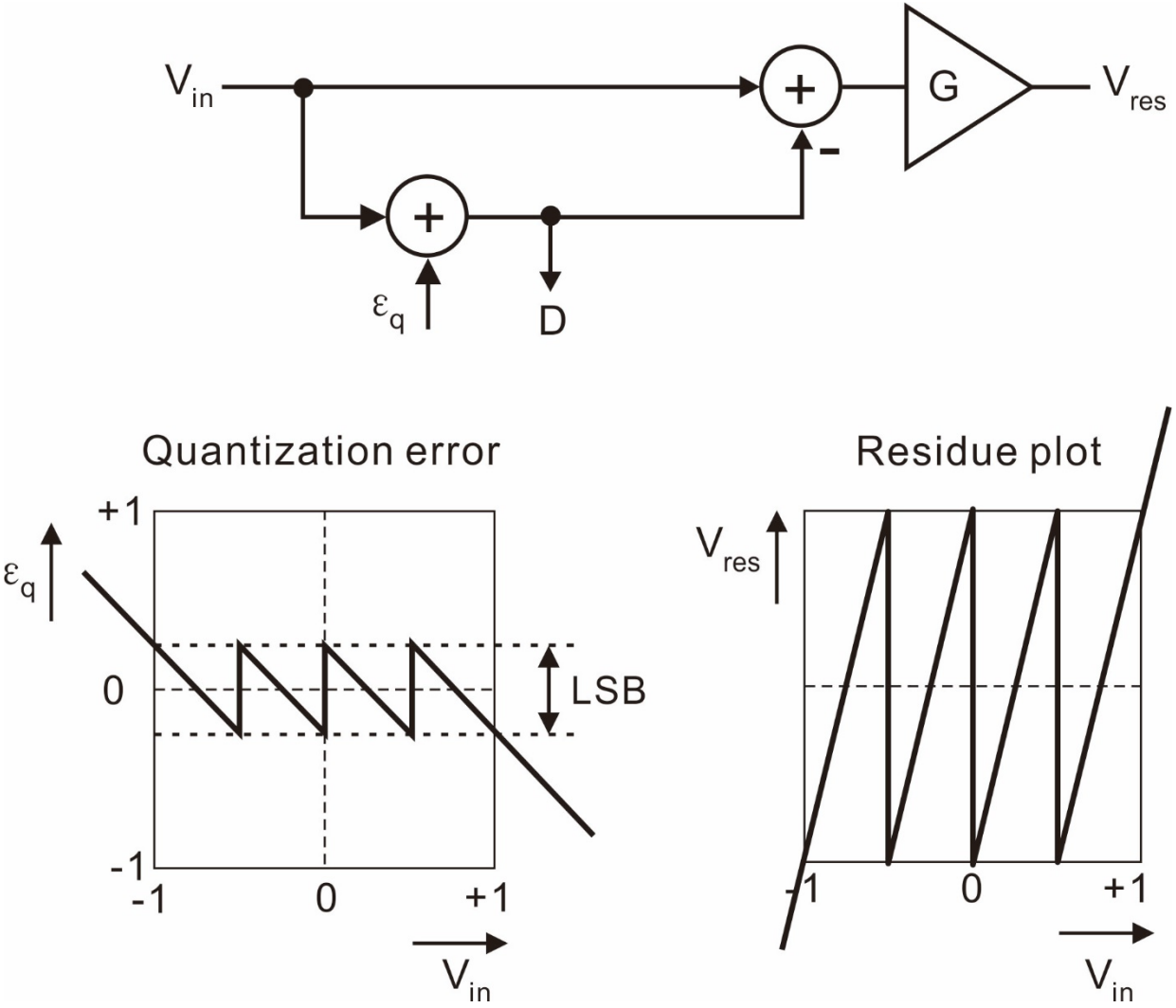
- Ignore timing/clock delays for simplicity



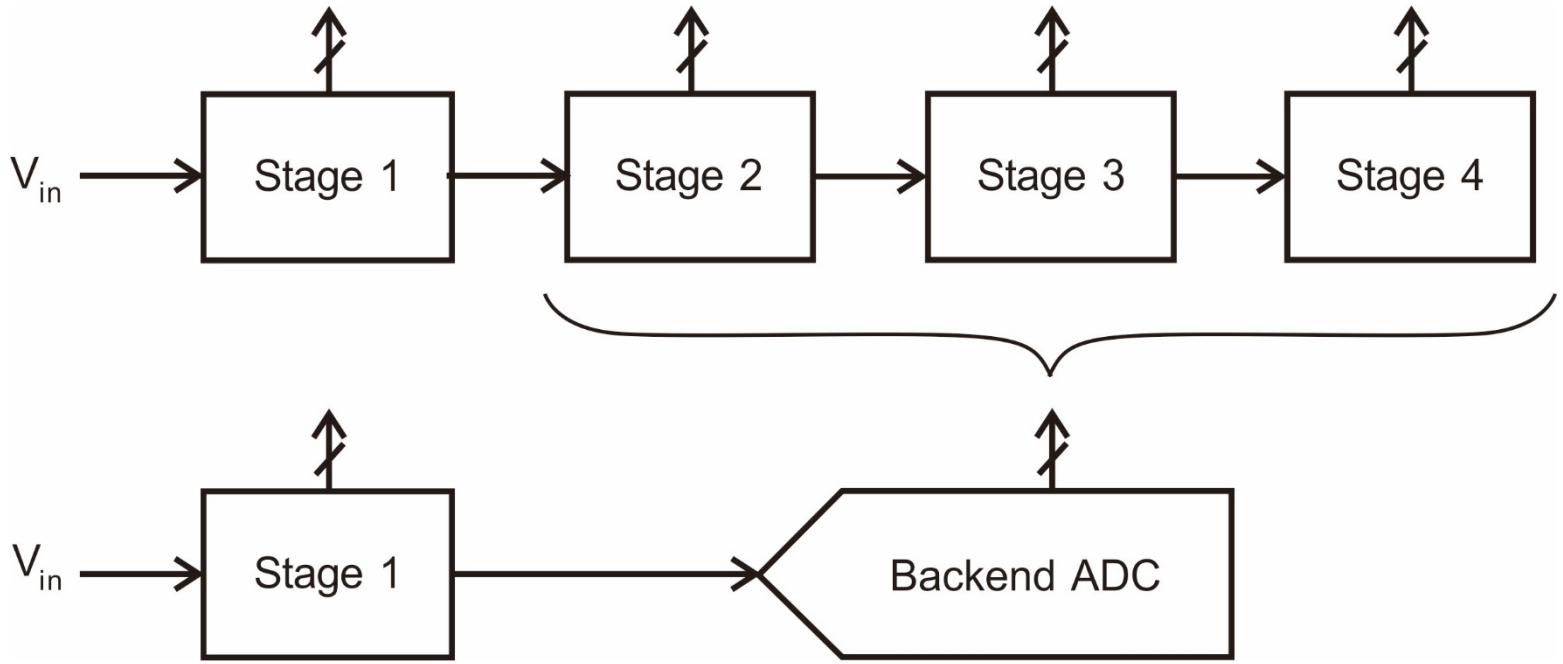
$$V_{res} = -G\varepsilon_q$$

$$D = V_{in} + \varepsilon_q$$

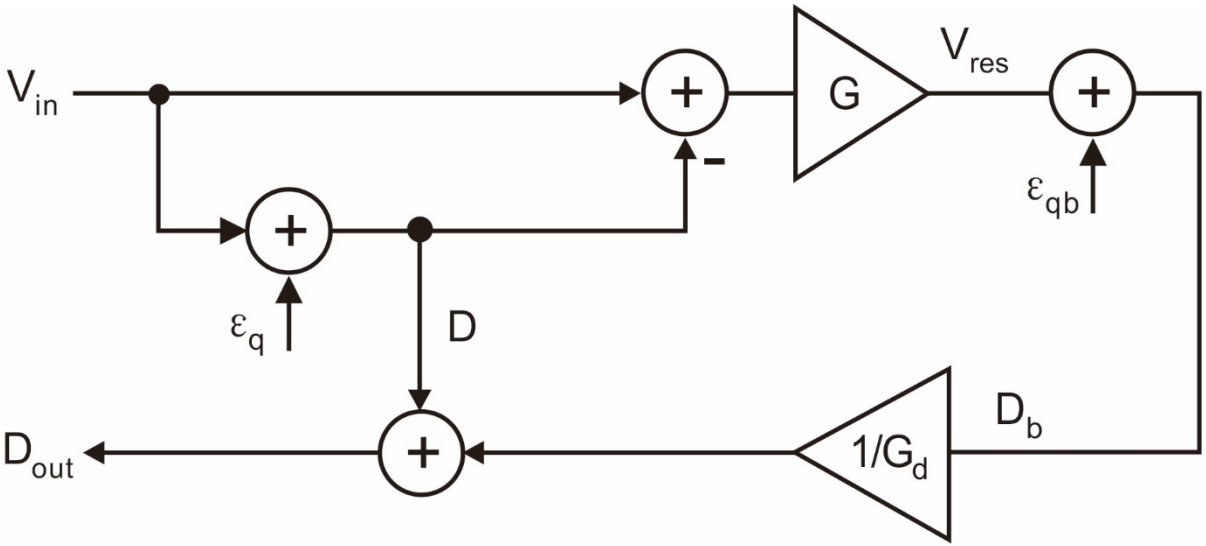
"Residue Plot" (2-bit Sub-ADC)



Pipeline Decomposition



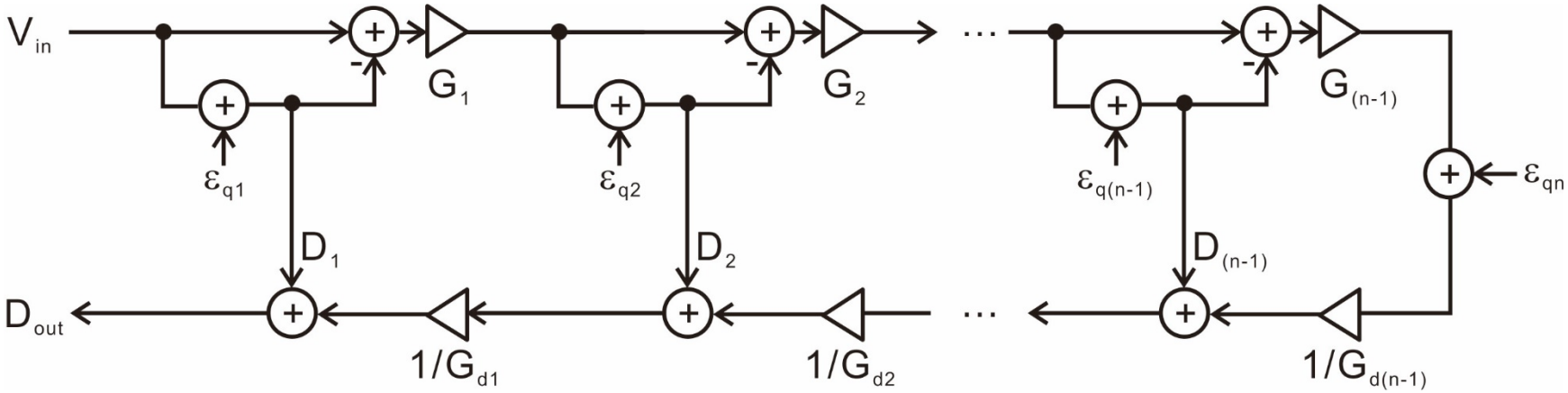
Resulting Model



$$D_{out} = V_{in} + \varepsilon_q\left(1 - \frac{G}{G_d}\right) + \frac{\varepsilon_{qb}}{G_d}$$

$$= V_{in} + \frac{\varepsilon_{qb}}{G_d} \qquad \text{for } G = G_d$$

Extension



$$D_{out} = V_{in} + \varepsilon_{q1}\left(1-\frac{G_1}{G_{d1}}\right) + \frac{\varepsilon_{q2}}{G_{d1}}\left(1-\frac{G_2}{G_{d2}}\right) + \ldots + \frac{\varepsilon_{q(n-1)}}{\prod_{j=1}^{n-2} G_{dj}}\left(1-\frac{G_{(n-1)}}{G_{d(n-1)}}\right) + \frac{\varepsilon_{qn}}{\prod_{j=1}^{n-1} G_{dj}}$$

- With ideal DACs and matched analog/digital gains ($G_{dj}=G_j$)

$$D_{out} = V_{in} + \frac{\varepsilon_{qn}}{\prod_{j=1}^{n-1} G_j} \quad \Rightarrow \quad B_{ADC} = B_n + \sum_{j=1}^{n-1} log_2 G_j$$

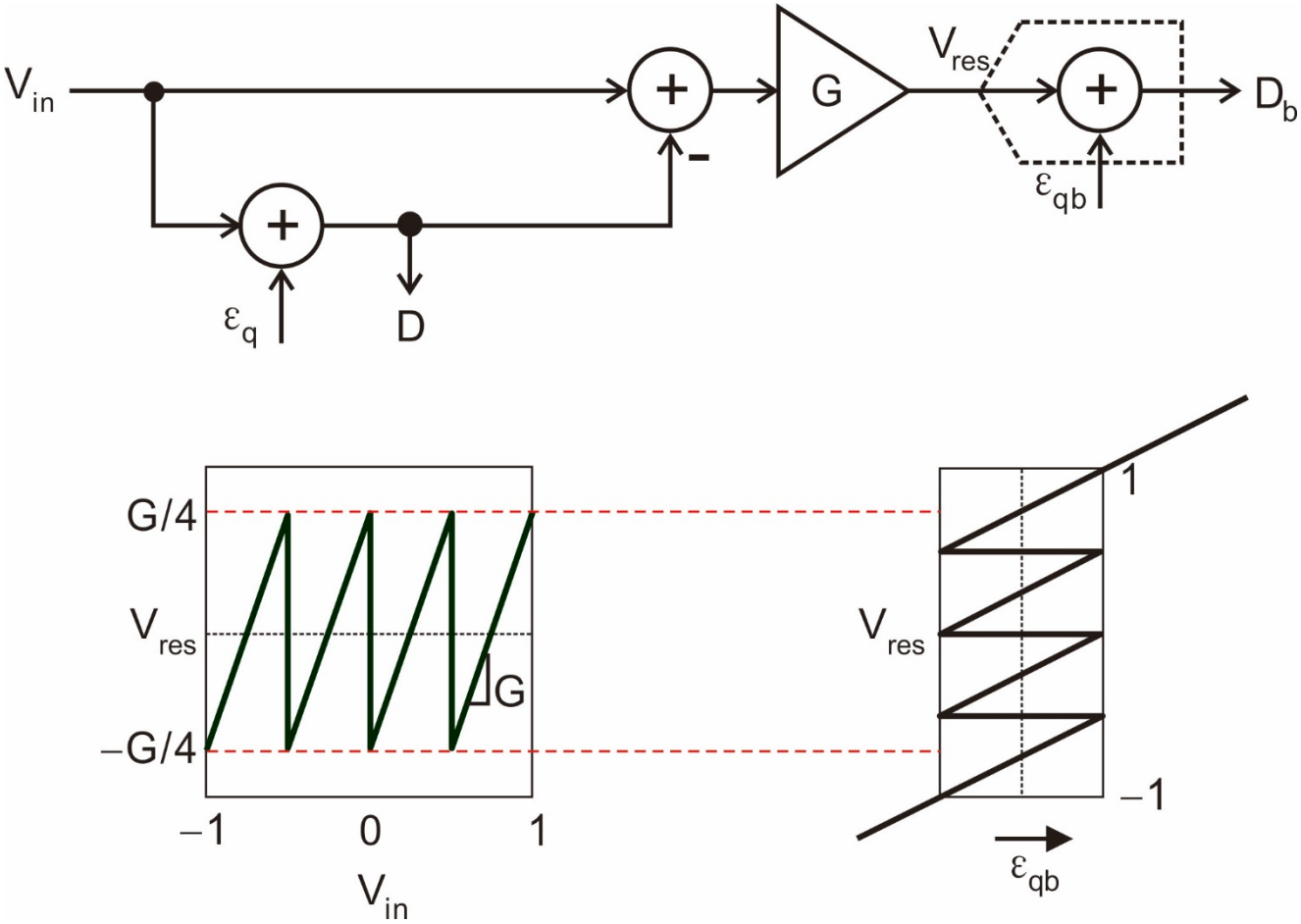
Analysis

- The only error in D_{out} is that of last quantizer, divided by aggregate gain
- Aggregate ADC resolution is independent of sub-ADC resolutions in stage 1 to stage (n-1).
- Gain is directly related to resolution
- Define "effective" resolution of the kth stage as $R_k=\log2(G_k)$
  - $B_{ADC} = R_1 + R_2 + R_3 + \ldots + R_{N-1} + B_N$

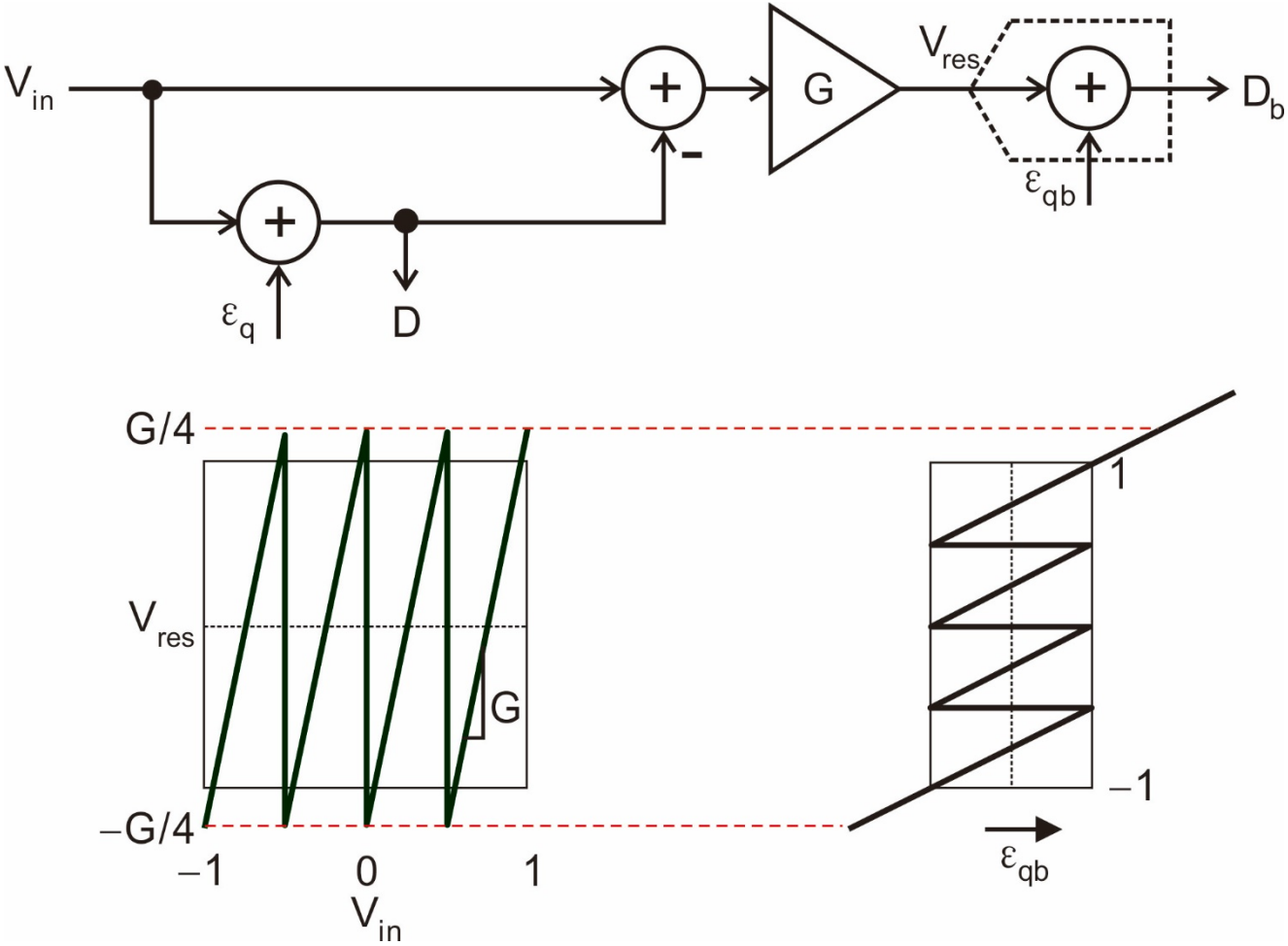
Questions

- How to pick inter-stage gain G for a given sub-ADC resolution?
- Nonidealities and solutions
  - Sub-ADC errors
  - Amplifier offset
  - Amplifier gain error
  - Sub-DAC error

Upper Bound for Stage Gain

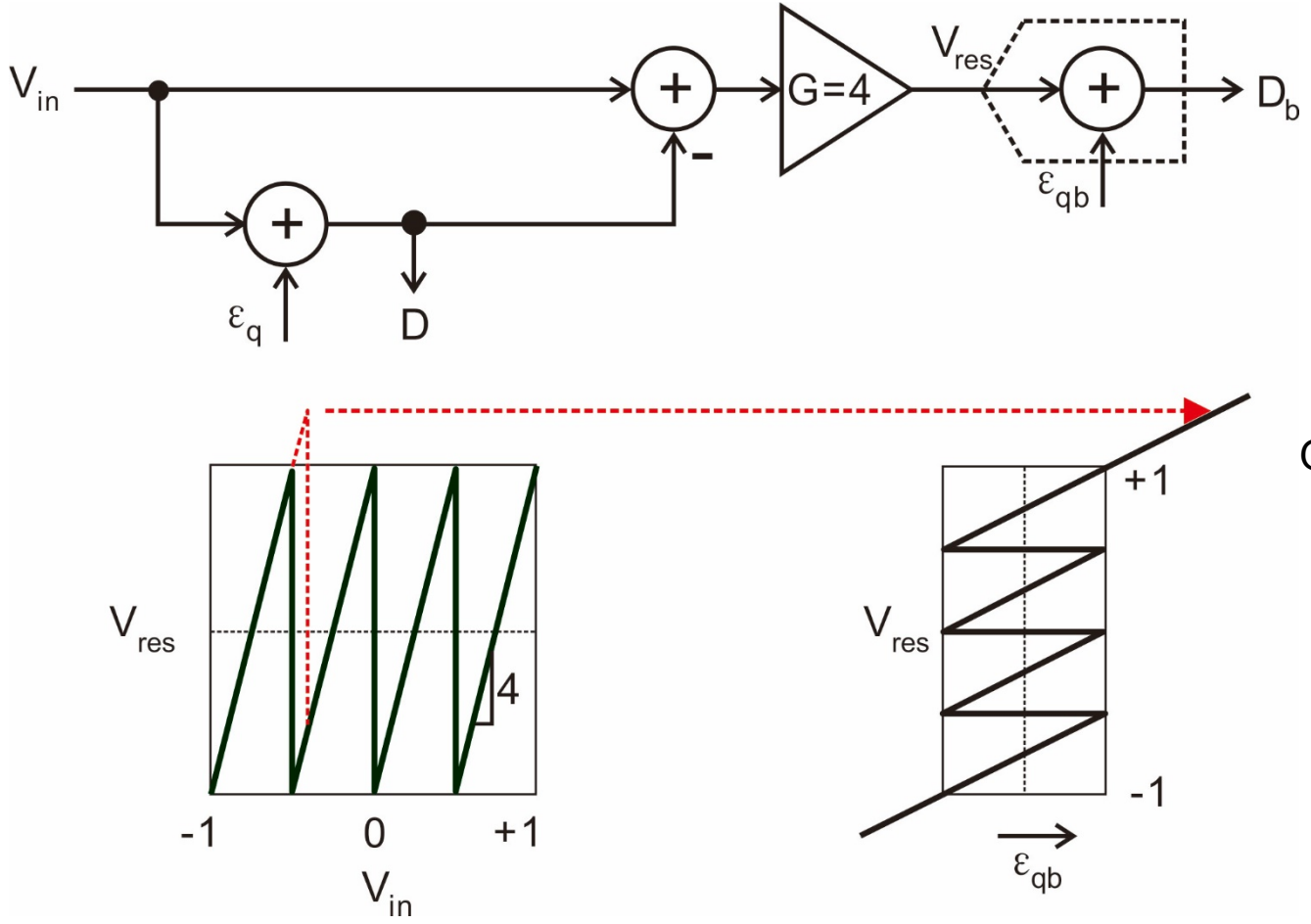


Upper Bound for Stage Gain



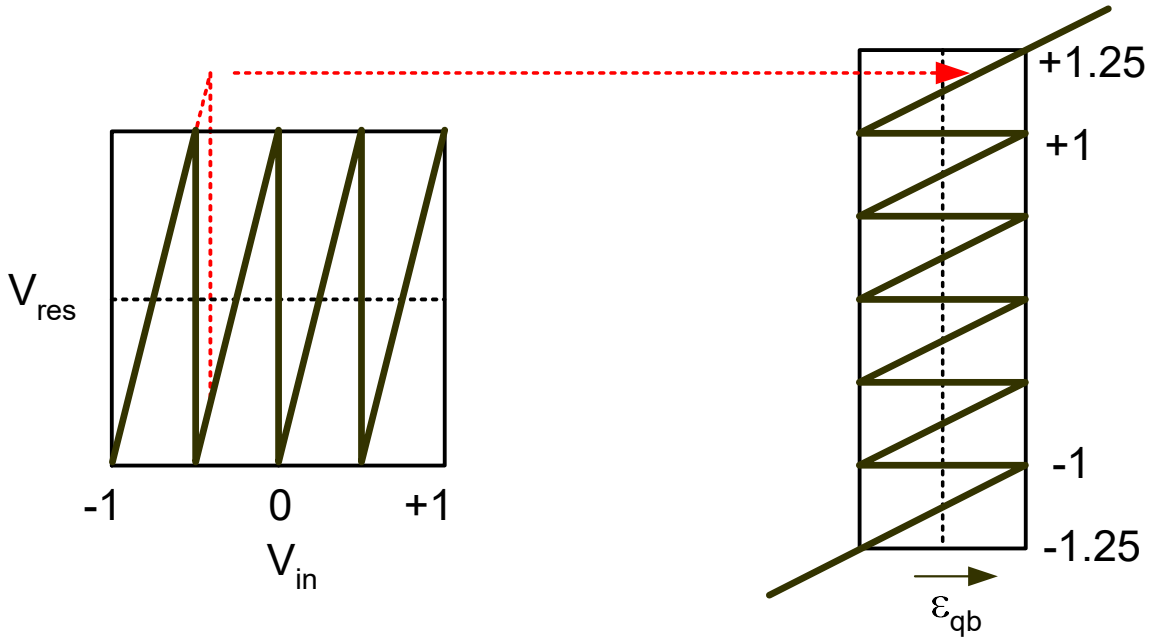
- Backend ADC will be overloaded for $G>2^B$

Issue with $G=2^B$



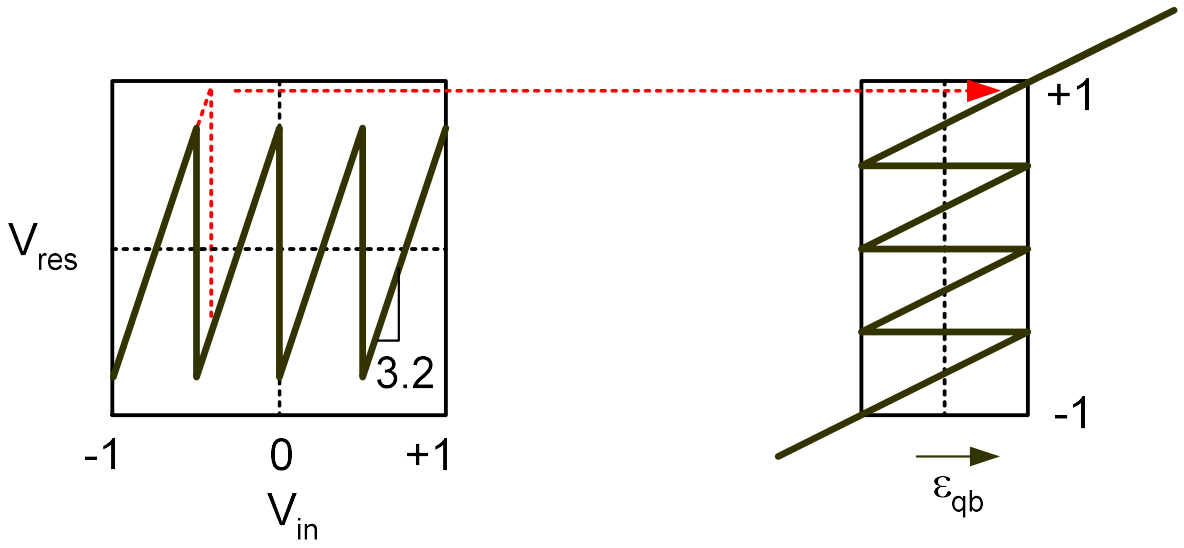
- Any error in sub-ADC decision levels (e.g. comparator offsets) will overload the backend ADC

Idea #1: G=2^B, extended backend



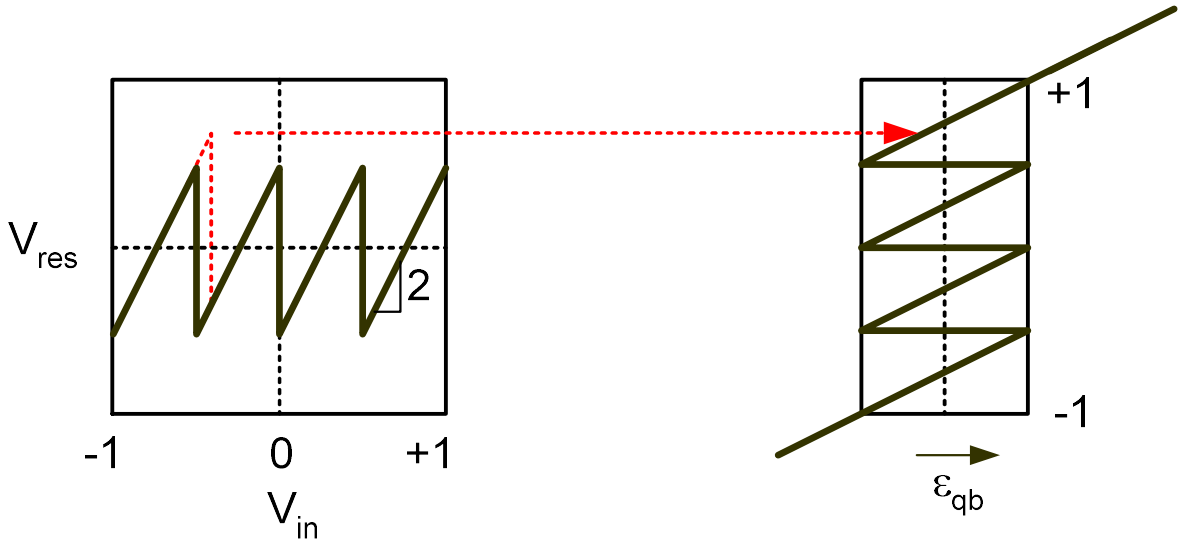
- Add extra decision levels in the backend stage
- See e.g. [Opris 1998]

Idea #2: G slightly less than 2^B



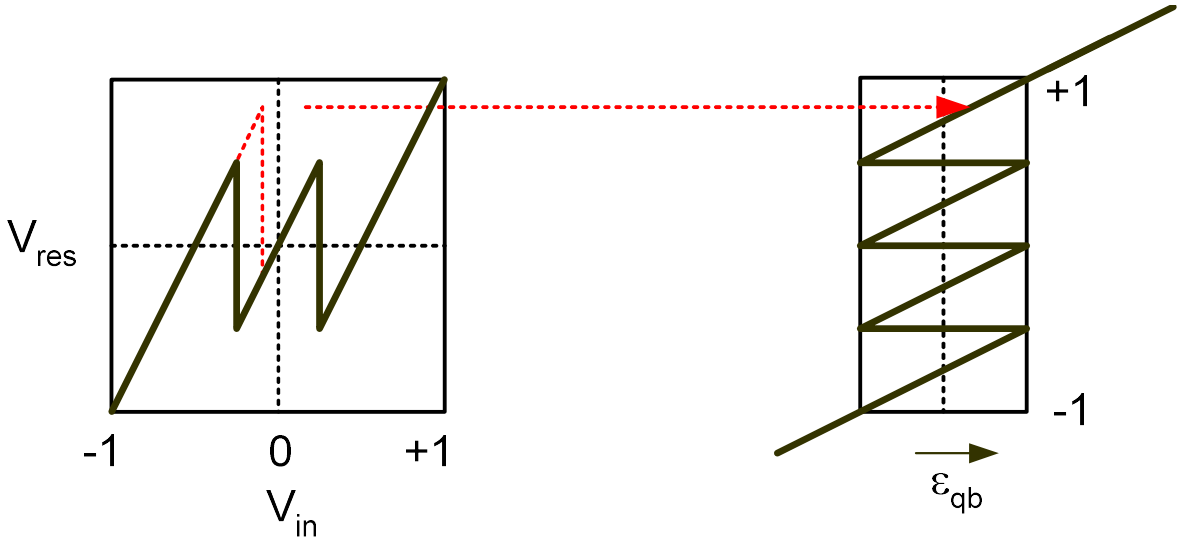
- Effective stage resolution can be non-integer ($R=\log_2 G$)
  - E.g. $R = \log_2 3.2 = 1.68$ bits
- See e.g. [Karanicolas 1993]

Idea #3: G < 2^B, but Power of Two



- Effective stage resolution is an integer
  - E.g. R = $\log_2 2$ = 1 = B-1
  - Digital hardware requires only a few adders, no need to implement fractional weights
- See e.g. [Mehr 2000]

Variant of Idea #3: "1.5-bit stage"

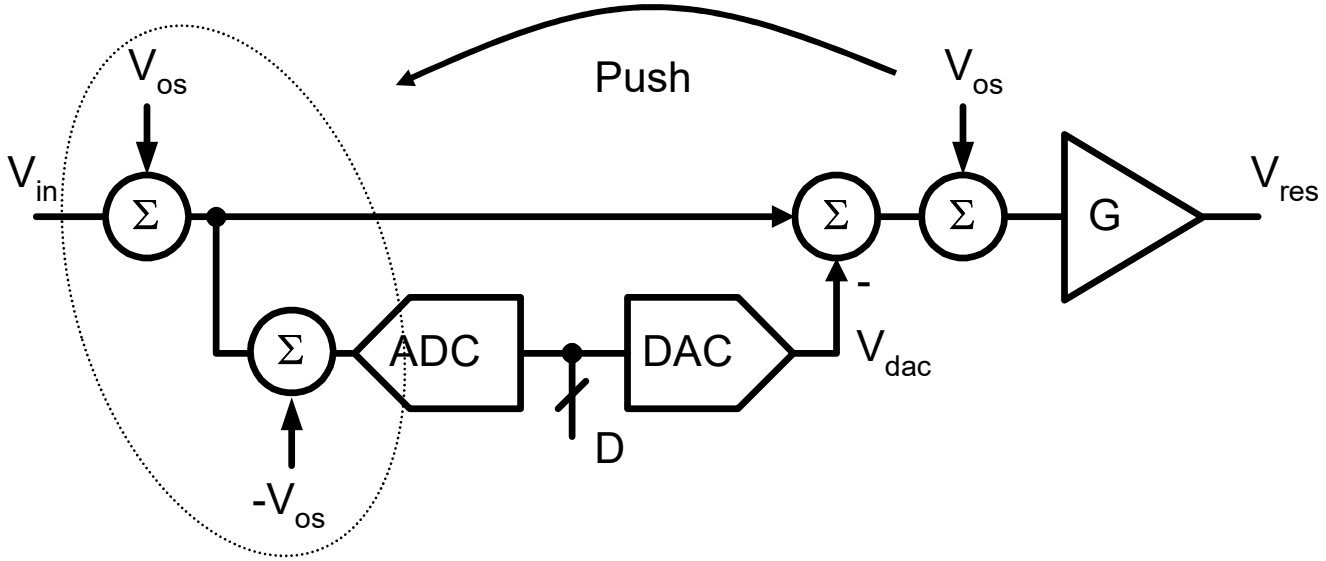


- Number of comparators reduced to 2
- DAC becomes easier to design
- R = $\log_2 2$ = 1 (effective stage resolution)
- See e.g. [Lewis 1992]

Summary on Sub-ADC Redundancy

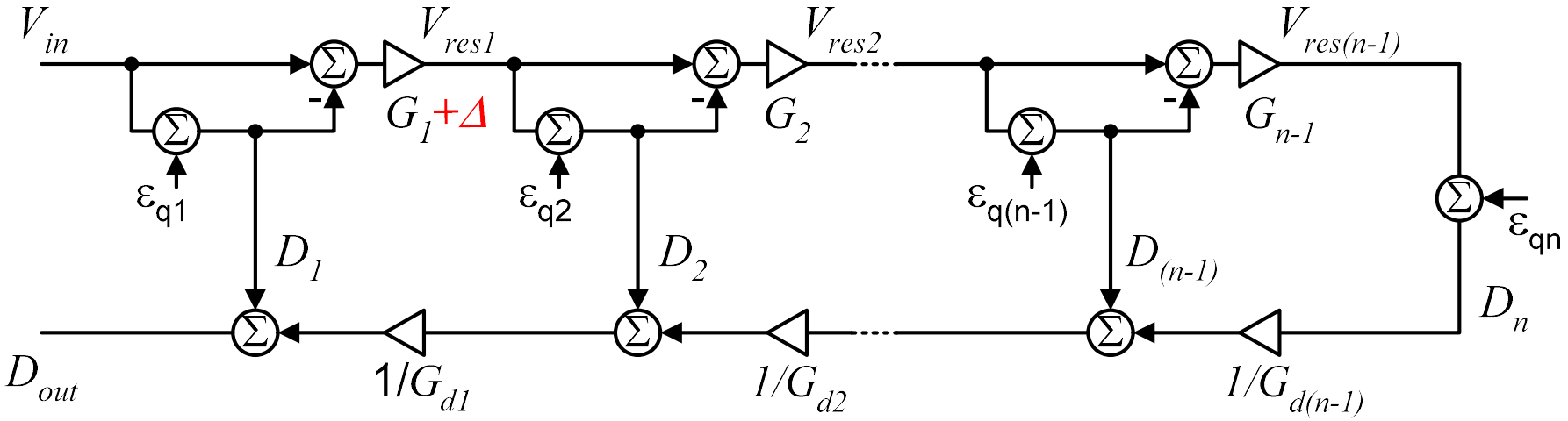
- We can tolerate sub-ADC errors as long as
  - The residue stays "inside the box", or
  - Another stage downstream returns the residue "into the box" before it reaches last quantizer
- This result applies to any stage in an n-stage pipelined ADC
  - Can always decompose pipeline into single stage + backend ADC

Amplifier Offset



- Amplifier offset can be referred toward stage input and results in
  - Global offset
    - Usually not a problem
  - Sub-ADC offset
    - Easily accommodated through redundancy

Gain Errors



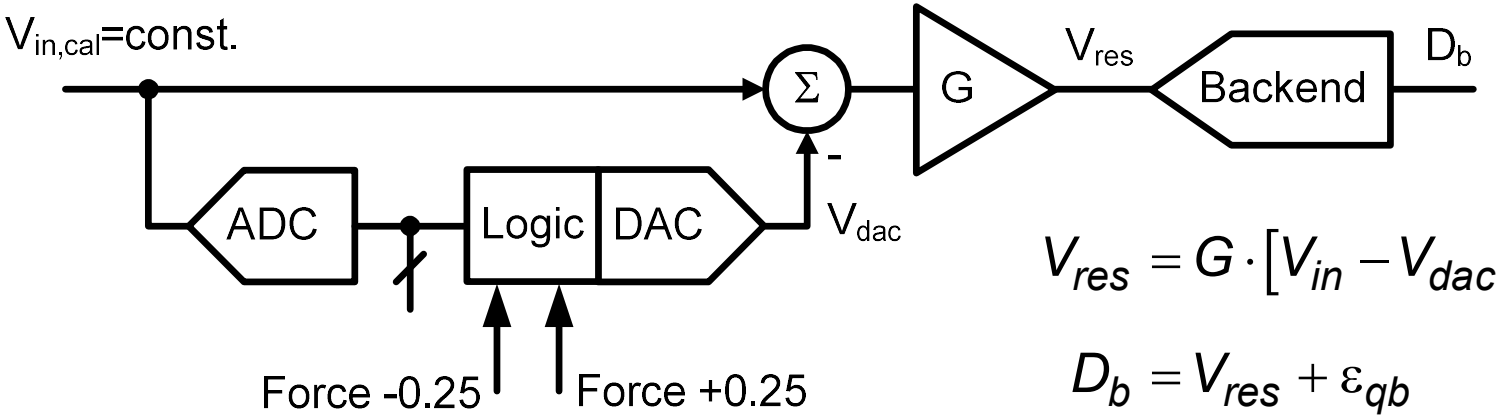
$$D_{out} = V_{in} + \varepsilon_{q1}\left(1 - \frac{G_1 + \Delta}{G_{d1}}\right) + \ldots + \frac{\varepsilon_{qn}}{\prod_{j=1}^{n-1} G_{dj}}$$

- Want to make $G_{d1} = G_1 + \Delta$

Digital Gain Calibration (1)

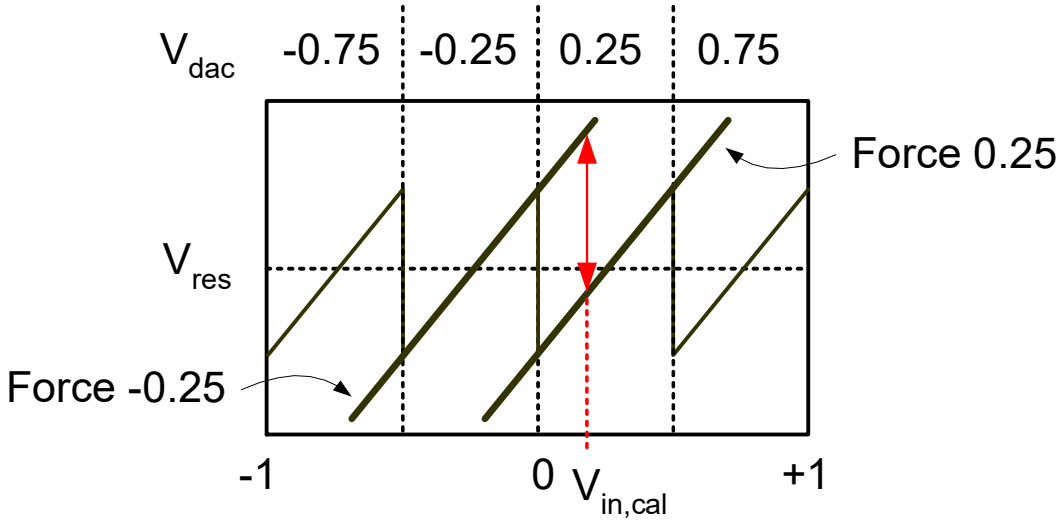
- Key idea
  - Analog gain error is not a problem as long as digital gain matches analog gain
  - Need to measure analog gain precisely
    - Use the backend

Digital Gain Calibration (2)



$$V_{res} = G \cdot [V_{in} - V_{dac}]$$

$$D_b = V_{res} + \varepsilon_{qb}$$



Digital Gain Calibration (3)

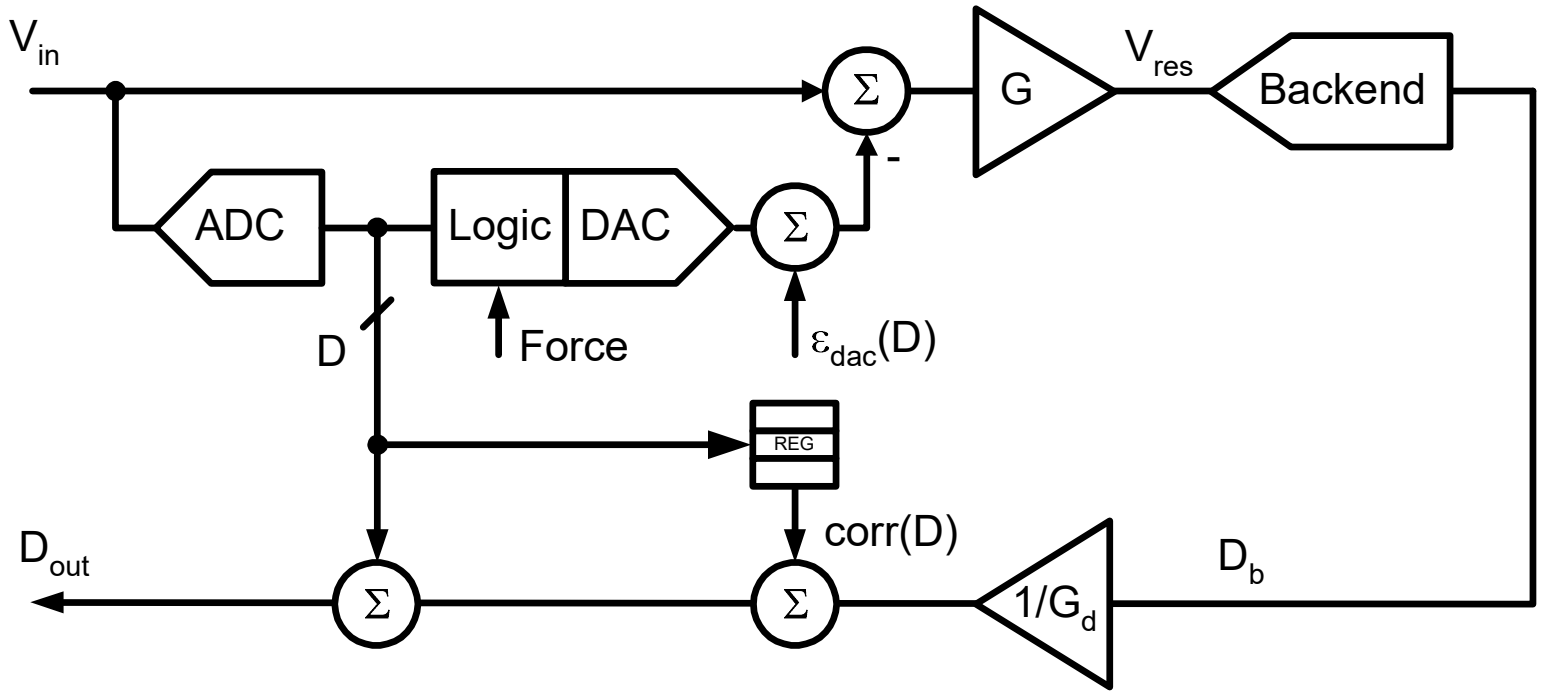
Step1: $$D_b^{(1)} = G \cdot \left[V_{in} + 0.25\right] + \varepsilon_{qb}^{(1)}$$

Step2: $$D_b^{(2)} = G \cdot \left[V_{in} - 0.25\right] + \varepsilon_{qb}^{(2)}$$

$$D_b^{(1)} - D_b^{(2)} = 0.5 \cdot G + \varepsilon_{qb}^{(1)} - \varepsilon_{qb}^{(2)}$$

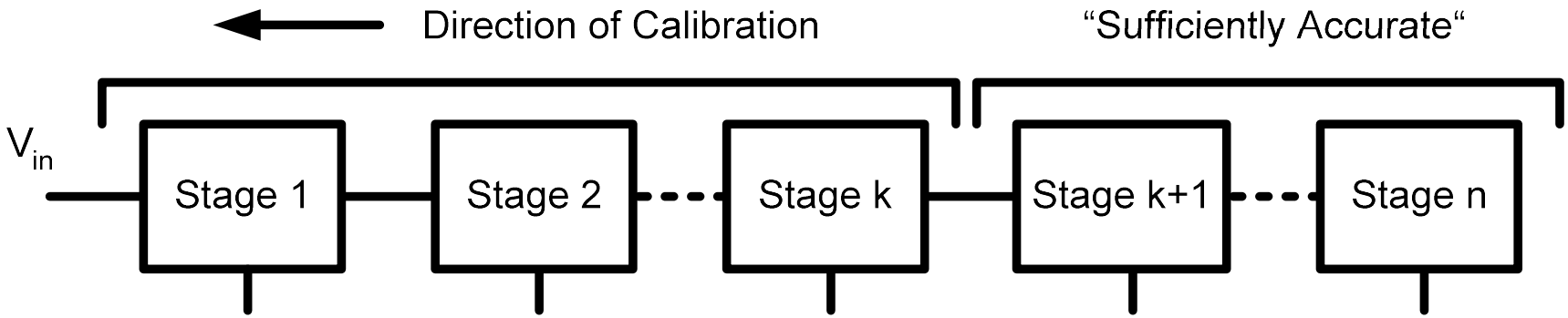
- We can minimize impact of quantization error by
  - Averaging (thermal noise dither)
  - Adding extra backend resolution
- Gain nonlinearity limits ENOB to about 10 bits.

DAC Calibration



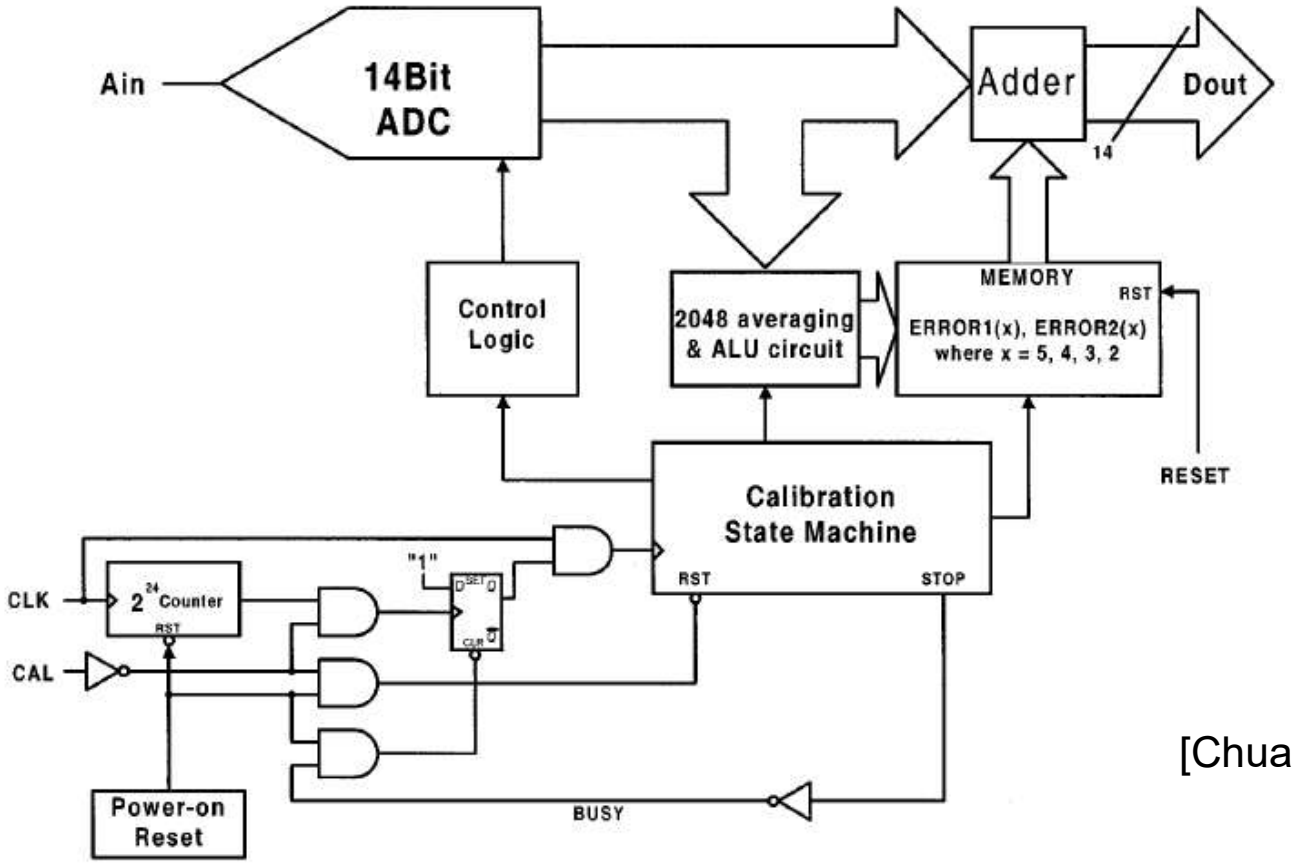
- Same concept as gain calibration
  - Sweep DAC codes and use backend to measure errors
- Store coefficients for each DAC transition in a look-up table
- What if the backend is not accurate?

Recursive Stage Calibration



- First few stages have most stringent accuracy requirements
  - Errors of later stages are attenuated by aggregate gain
- Commonly used algorithm [Karanicolas 1993]
  - Take ADC offline
  - First calibrate the least significant stage that needs calibration
  - Move to the next significant stage and continue toward stage 1

Calibration Hardware Example



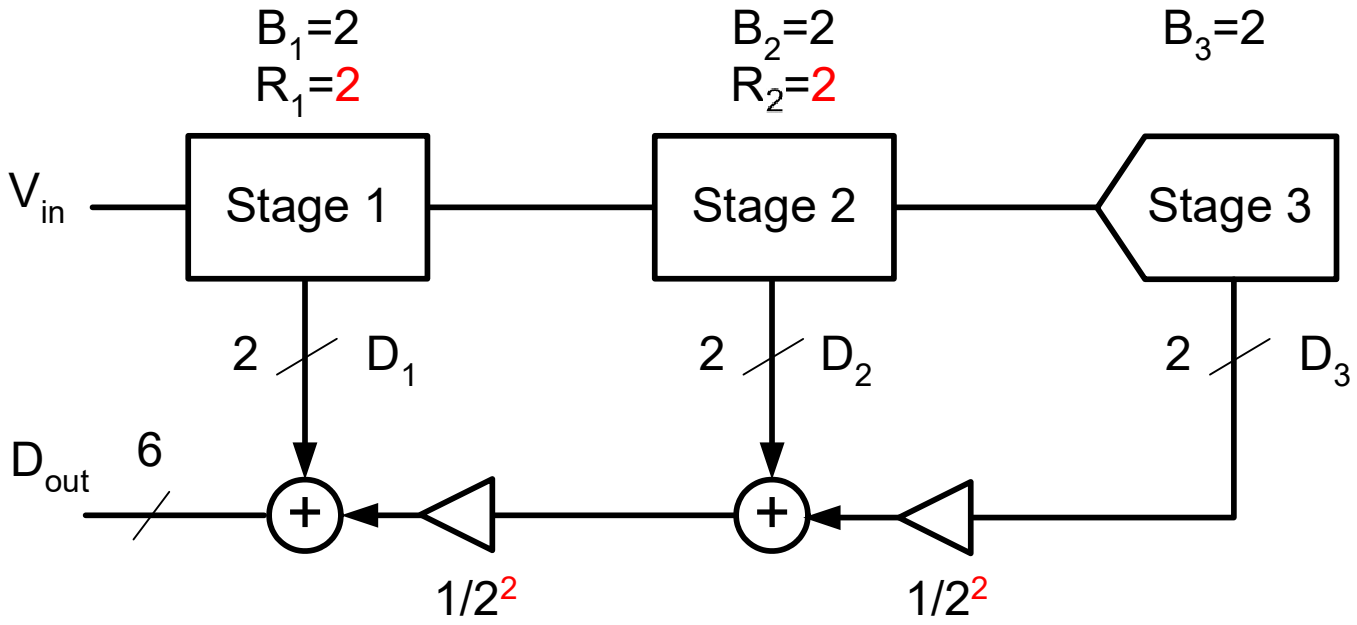
[Chuang 2002]

Alternative Schemes

- Other foreground calibration schemes
  - Calibrate ADC starting from first stage [Singer 2000]
  - Connect stages in a circular loop [Soenen 1995]
- Foreground calibration is not accurate enough when errors drift
  - Gain error drifts due to temperature variation and device aging
  - DAC error due to capacitor mismatch does not drift
- Background calibration techniques [e.g., Ming 2001]
  - Can keep track of gain variations ☺
  - Without disturbing the ADC normal operation ☺

Combining the Bits (1)

- Example1: Three 2-bit stages, no redundancy

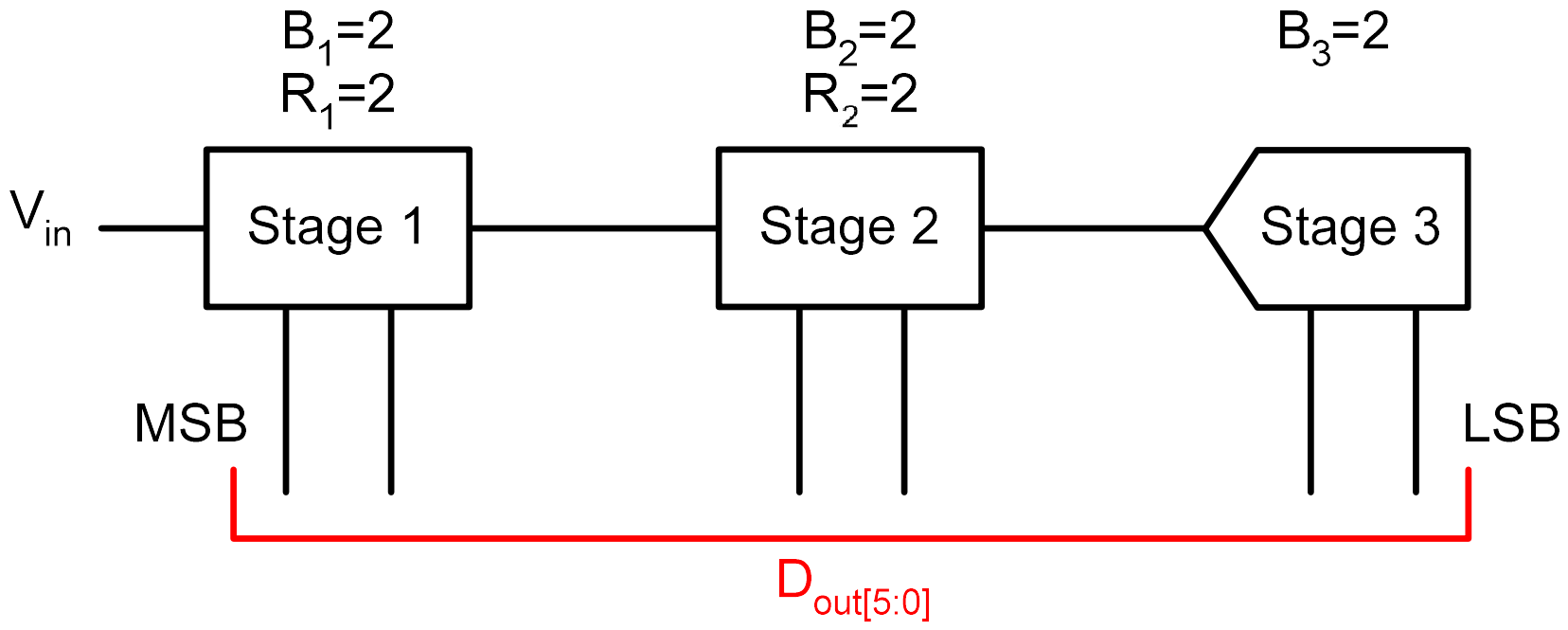


$$D_{out} = D_1 + \frac{1}{4}D_2 + \frac{1}{16}D_3$$

Combining the Bits (2)

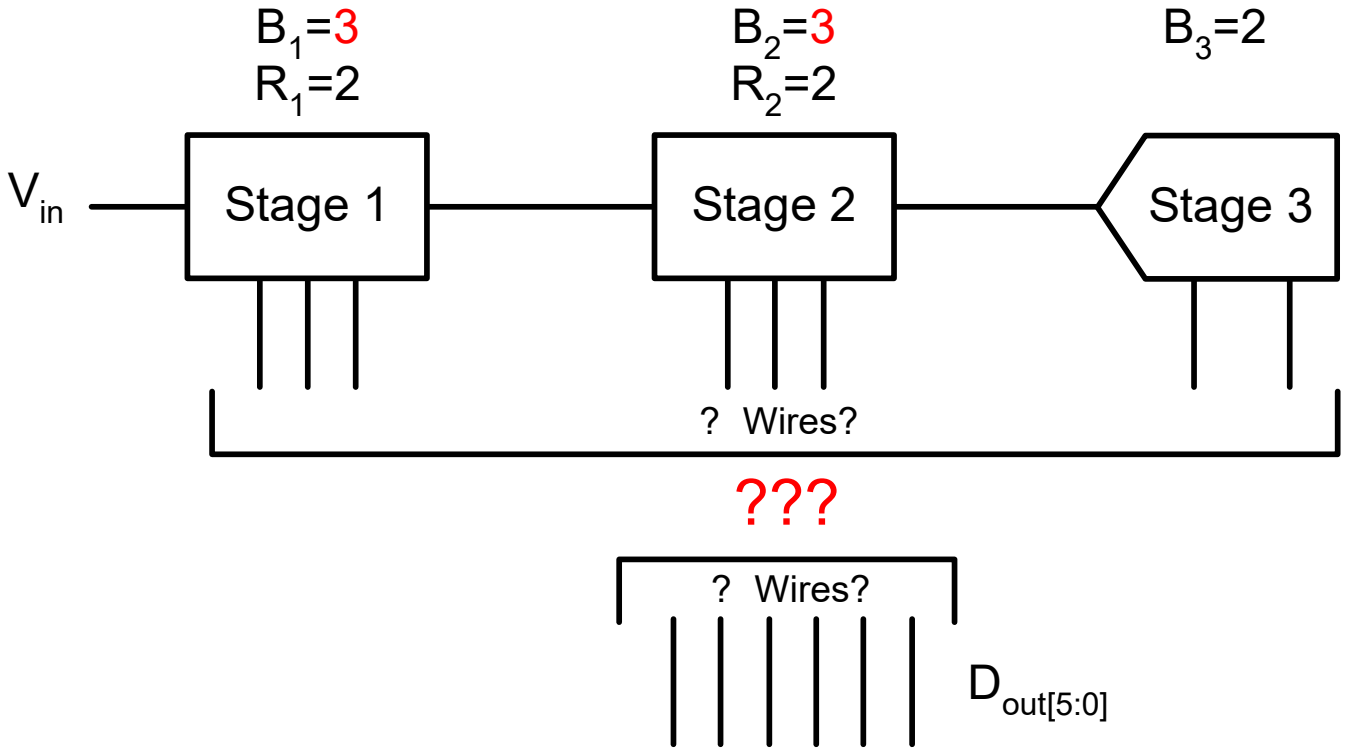
```
D1      XX
D2         XX
D3            XX
------------
Dout    XXXXXX
```

- Only bit alignment
- No digital circuits needed



Combining the Bits (3)

- Example2: 1 bit redundancy in stages 1 and 2

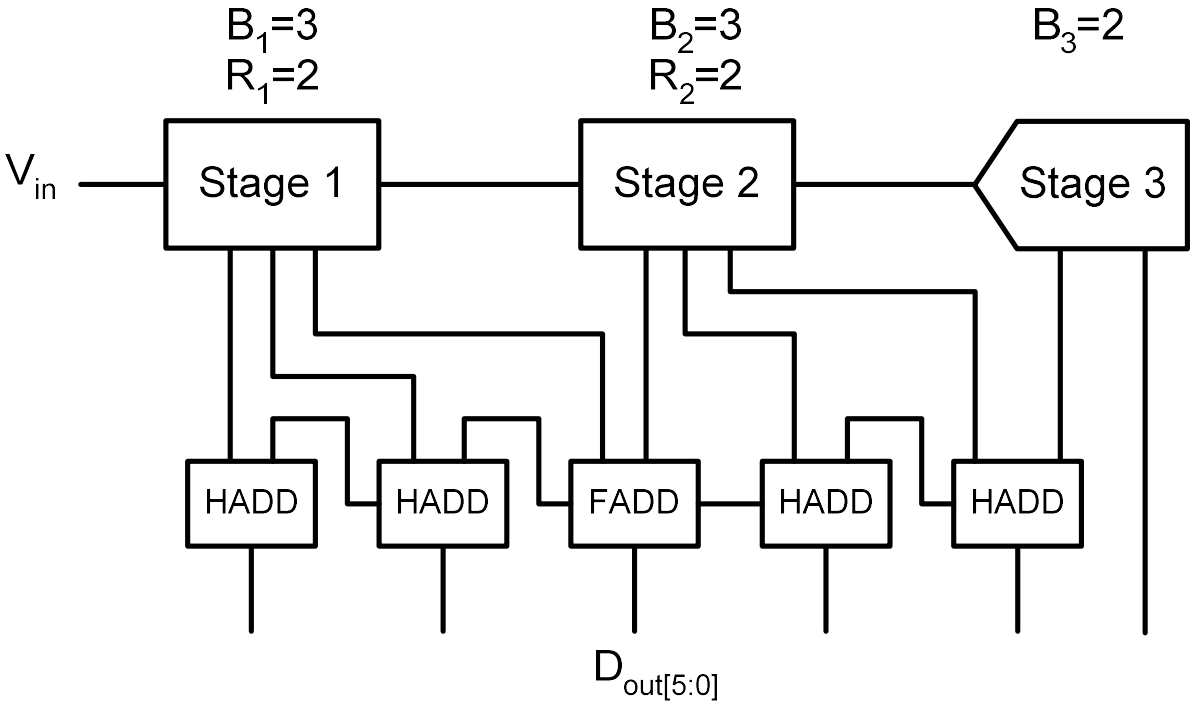


Combining the Bits (4)

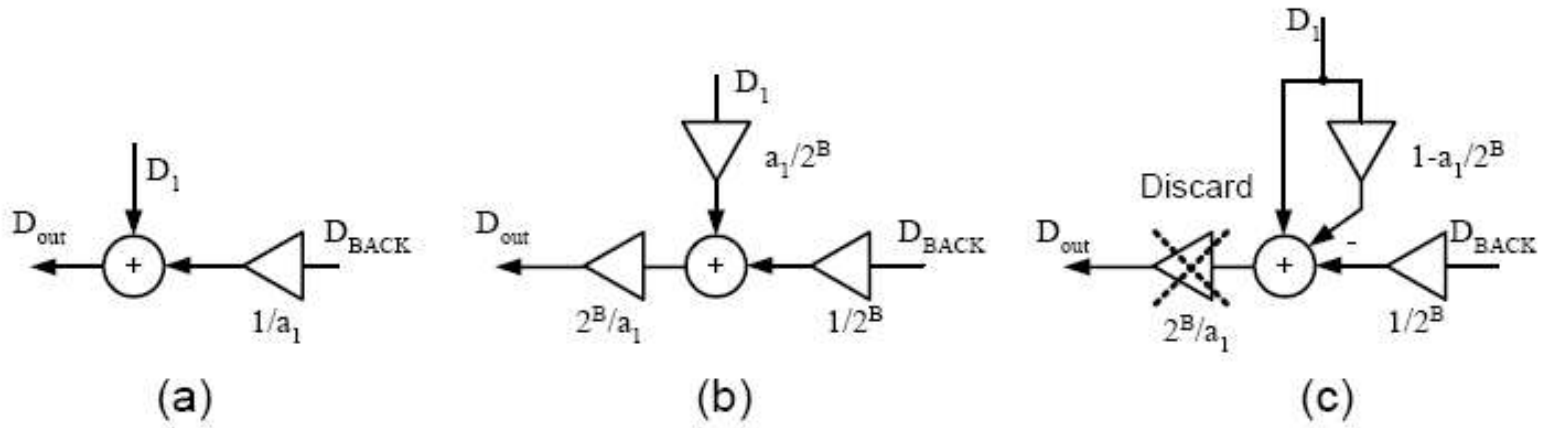
$$D_{out} = D_1 + \frac{1}{4}D_2 + \frac{1}{16}D_3$$

```
D1      XXX
D2        XXX
D3          XX
------------
Dout    DDDDDD
```

- Bits overlap
- Need adders



Combining the Bits (5)



- For fractional weights (e.g. radix <2), avoid the use of complex multipliers
  - 10X = 8X + 2X
- See e.g. [Karanicolas 1993]